

Asymptotically Optimal Mixed-State Cloning

Abstract

We determine the asymptotic global fidelity for cloning full-rank nondegenerate finite-dimensional mixed states in the proportional regime $m_n/n \rightarrow \gamma > 1$. For the unitary orbit of a known spectrum $p_1 > \dots > p_d > 0$, the optimal value is

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} \frac{(1 - q_{ij})(1 - q_{ij}^{(\gamma)})}{(1 - \sqrt{q_{ij} q_{ij}^{(\gamma)}})^2}, \quad q_{ij} = p_j/p_i, \quad q_{ij}^{(\gamma)} = 1 - \frac{1 - q_{ij}}{\gamma}.$$

It is achieved by a Schur–Weyl protocol that measures the Young diagram and applies a Cartan/Parthasarathy–Ranga Rao–Varadarajan (PRV) sector map between typical Weyl irreducible representations. The converse follows from finite-dimensional local asymptotic normality and optimal displaced-thermal Gaussian amplification. When the spectrum is unknown, the same protocol becomes adaptive by choosing the output Young diagram as a rounded dilation of the measured input diagram; the optimal local asymptotic fidelity is

$$F_{\text{univ}}(\gamma, p) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{d-1} F_{\text{orb}}(\gamma, p),$$

where the prefactor is the optimal classical Gaussian amplification fidelity for the spectral LAN component. Thus, in the full-rank nondegenerate regime, the finite-dimensional mixed-state cloning problem admits a sharp asymptotic solution for global all-site fidelity.

1 Introduction

The no-cloning theorem says that an arbitrary unknown quantum state cannot be copied perfectly [WZ82, Die82]. Approximate cloning asks instead for the best physical channel which maps n identical input systems into m output systems whose joint state is as close as possible to m ideal copies. For pure states, this problem is well understood: after the first universal qubit machines, Werner gave a covariant construction that solves universal pure-state cloning in arbitrary finite dimension with global fidelity equal to a symmetric-subspace dimension ratio [BH96, GM97, BDE+98, Wer98, KW99]. Broader cloning variants, reviews, and asymptotic/state-estimation connections are treated in [MP95, BEM98, SIGA05, FWJ+14, BA06, CD06, CY14]. We recall this construction and its global and one-site fidelities in Section 2.2. In the decades since Werner’s theorem, no comparably general solution has been available for genuinely mixed finite-dimensional states with a global all-site product fidelity. The main contribution of this paper is to give such a solution in the asymptotic proportional-output regime, for full-rank nondegenerate states. [Possible referee-sensitive wording: “to our knowledge, the first sharp asymptotic solution for global all-site mixed-state cloning in arbitrary fixed finite dimension.”]

It is important to separate *cloning* from *broadcasting*. A cloning channel aims to approximate the full product state $\rho^{\otimes m}$. A broadcasting channel only asks that selected one-site marginals equal, or approximate, the input state; the joint output may be strongly correlated and far from

a product. The single-copy no-broadcasting theorem says that a set of states can be broadcast exactly only when the states commute [BCFJS96]. This theorem should not be confused with multi-input broadcasting: with several input copies, the obstruction is different. In particular, D’Ariano–Macchiavello–Perinotti showed for mixed qubit states that the no-broadcasting theorem does not extend directly beyond a single input copy, and that for four or more input copies one can even increase the local purity while broadcasting, a phenomenon they called superbroadcasting [DMP05]. Subsequent work developed universal and phase-covariant superbroadcasting and optimal mixed-qubit broadcasting maps [BDMP05, BDMP06, DF07]. In regimes where the one-site output is more pure than the input, exact marginal broadcasting is obtained by adding local depolarizing noise; this still does not produce independent product copies. For displaced thermal oscillator states, D’Ariano–Perinotti–Sacchi also give an explicit exact broadcasting construction when the input thermal photon number is known [DPS06]. These results are about marginal outputs, not global product fidelity. They therefore do not solve the cloning problem studied here.

Much of the previous finite-dimensional approximate mixed-state cloning and broadcasting literature focuses on qubits, special covariance classes, purification, partial cloning, or marginal figures of merit. Cirac–Ekert–Macchiavello introduced a decomposition of $\rho^{\otimes N}$ for optimal single-qubit purification and noted its connection with optimal cloning and estimation of mixed states [CEM99]. Rastegin developed global-fidelity, relative-error, and state-dependent bounds for mixed-state cloning [Ras03a, Ras02, Ras03c]; Fan constructed cloners for mixed states supported on symmetric subspaces and for identical mixed qubits [Fan03, FLS07]; and Kazakov studied partial cloning and mixed-state cloning in a related finite-dimensional setting [Kaz07]. The LAN approach has also been used for asymptotic purification and dilution of mixed qubits [BGA11]. Continuous-variable cloning provides another comparison point: optimal coherent-state cloners are realized by linear amplification and beam splitters [CIR00, BCI+01], and Guta–Matsumoto solved optimal cloning of displaced mixed Gaussian states by reducing it to least-noise amplification [GM06]. Our converse uses precisely this Gaussian amplification theorem after finite-dimensional local asymptotic normality reduces the mixed-state orbit to independent oscillator modes.

We consider two finite-dimensional regimes. In the known-spectrum problem, the spectrum $p = (p_1, \dots, p_d)$, $p_1 > \dots > p_d > 0$, is fixed and known, and one clones the unitary orbit

$$\rho_{p,g} = U(g) \operatorname{diag}(p) U(g)^*.$$

Only the eigenbasis parameter is unknown. In the unknown-spectrum problem, the spectrum also varies locally; the Schur–Weyl Young label then carries a classical Gaussian LAN component. The known-spectrum model contains only the off-diagonal quantum oscillator modes, whereas the unknown-spectrum model contains both the classical spectral Gaussian part and the quantum orbit part.

The abbreviation PRV refers to the theorem of Parthasarathy, Ranga Rao, and Varadarajan on extremal components in tensor products of irreducible representations. We use its Cartan-component special case: if α and β are dominant weights, then $\mathcal{H}_{\alpha+\beta}$ appears canonically in $\mathcal{H}_\alpha \otimes \mathcal{H}_\beta$. The associated normalized inclusion is the PRV inclusion. Our sector channel is the adjoint, trace-normalized operation obtained by embedding an output irrep $\mathcal{H}_\nu$ into $\mathcal{H}_\mu \otimes \mathcal{H}_{\nu-\mu}$ and tracing against the auxiliary factor. A formal definition is given in Section 2.4 before the protocol is stated.

We use the global, all-site fidelity

$$F(\sigma, \tau) = \|\sqrt{\sigma}\sqrt{\tau}\|_1^2$$

between the actual m -partite output and the ideal state $\rho^{\otimes m}$. This is stricter than one-site marginal fidelity and is the key distinction from the broadcasting/superbroadcasting literature. We work in

the proportional-output regime

$$\frac{m_n}{n} \longrightarrow \gamma > 1.$$

1.1 Informal statement of the main results

Let

$$\rho_p = \text{diag}(p_1, \dots, p_d), \quad p_1 > \dots > p_d > 0,$$

and let $\rho_{p,g} = U(g)\rho_p U(g)^*$. For every positive root $i < j$, set

$$q_{ij} = \frac{p_j}{p_i}, \quad q_{ij}^{(\gamma)} = 1 - \frac{1 - q_{ij}}{\gamma}.$$

The known-spectrum answer is

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} \frac{(1 - q_{ij})(1 - q_{ij}^{(\gamma)})}{(1 - \sqrt{q_{ij}q_{ij}^{(\gamma)}})^2}. \quad (1)$$

Equivalently,

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} \left(\frac{\sqrt{\gamma} + \sqrt{q_{ij}(\gamma - 1 + q_{ij})}}{\gamma + q_{ij}} \right)^2. \quad (2)$$

The Schur–PRV protocol achieves (1). Conversely, any global channel that beat (1) would, through the Kahn–Guta LAN maps, induce a displaced-thermal Gaussian cloner beating the least-noise Gaussian amplifier. That contradicts the Gaussian cloning theorem of Guta–Matsumoto together with the fidelity optimality argument given below [KG09, GM06].

If the spectrum is unknown, the optimal local asymptotic fidelity at a fixed nondegenerate spectrum p becomes

$$F_{\text{univ}}(\gamma, p) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{d-1} F_{\text{orb}}(\gamma, p). \quad (3)$$

The extra factor is purely classical. It is the squared Hellinger fidelity between the output Young-label Gaussian produced by optimally amplifying the measured Schur label and the target Young-label Gaussian. Thus measuring the Young diagram and adapting the PRV sector map is optimal, but it cannot remove the $n^{-1/2}$ -scale uncertainty in the spectrum unless the spectrum is given as side information.

1.2 Related work and comparison

The closest existing finite-dimensional all-site construction is the random-purification plus pure-state-cloning protocol of Li–Theil–Harrow–Chuang [LTHC26a]. It is universal over rank- r mixed states: one first purifies the input by a random-purification channel, applies Werner’s optimal pure-state cloner in dimension $D = dr$, and traces out the purifying registers. Thus its stated all-site fidelity guarantee is precisely the Werner global-fidelity dimension ratio (10) with $D = dr$:

$$F_{\text{RP}}(n, m; d, r) \geq \frac{\binom{n+dr-1}{dr-1}}{\binom{m+dr-1}{dr-1}}, \quad (4)$$

so, for $m/n \rightarrow \gamma$,

$$F_{\text{RP}}^\infty(\gamma; d, r) \geq \gamma^{-(dr-1)}.$$

For full rank $r = d$, this is $\gamma^{-(d^2-1)}$. In contrast, our fixed-spectrum orbit fidelity obeys

$$F_{\text{orb}}(\gamma, p) > \gamma^{-d(d-1)/2},$$

with strict inequality for every $\gamma > 1$ and full-rank p . Even in the unknown-spectrum problem,

$$F_{\text{univ}}(\gamma, p) > \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^{d-1} \gamma^{-d(d-1)/2} > \gamma^{-(d^2-1)}.$$

This comparison should be read carefully: (4) is a universal rank- r guarantee for a different, harder task, not the exact fidelity of that channel on a fixed-spectrum orbit.

Other mixed-state cloning and broadcasting results address different operational criteria. Rastegin’s bounds apply to mixed-state cloning, relative-error criteria, and state-dependent cloning, but they do not identify the sharp arbitrary-dimensional proportional-output all-site asymptotics considered here [Ras03a, Ras02, Ras03c]. Fan’s mixed-state cloners treat symmetric-subspace inputs and identical mixed qubits, again with criteria different from our global product fidelity [Fan03, FLS07]; Kazakov’s partial-cloning construction is another qubit-level mixed-state copying model [Kaz07]. The superbroadcasting literature optimizes one-site marginals of correlated outputs; it demonstrates that marginal broadcasting can be exact, or even locally purifying, when multiple input copies are available, but it does not require the joint output to approximate $\rho^{\otimes m}$ [DMP05, BDMP06, DF07, DPS06]. Mixed Gaussian cloning, by contrast, is very close in spirit to our LAN limit; Guta–Matsumoto’s displaced-thermal amplifier theorem is the Gaussian converse used in this paper [GM06].

[Possible addition for final introduction: mention explicitly that the finite- n universal mixed-state analogue of Werner’s theorem remains open; our statement is the sharp asymptotic proportional-output solution for the full-rank nondegenerate finite-dimensional regime.]

1.3 Organization

Section 2 collects the finite-dimensional representation-theoretic background: fidelity, Werner’s pure-state cloner, Schur–Weyl decomposition, Young concentration, and the PRV sector channel. Section 3 introduces classical and quantum statistical models, bosonic Gaussian states, quantum-limited amplification, finite-dimensional local asymptotic normality, and the Gaussian cloning converse. Section 4 proves known-spectrum achievability and the global converse. Section 5 treats the adaptive unknown-spectrum protocol, the expanding-window local converse, and the compact strict-chamber global result. Section 6 compares with earlier cloning and broadcasting results, and Section 7 discusses extensions. The only technical proof deferred to the appendix is the hybrid classical–quantum Gaussian product converse used in the unknown-spectrum upper bound.

2 Representation-theoretic preliminaries

2.1 Fidelity conventions

We use squared Uhlmann fidelity

$$F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1^2$$

and root fidelity

$$B(\rho, \sigma) = \sqrt{F(\rho, \sigma)}.$$

For classical probability distributions P, Q , root fidelity is the Hellinger affinity

$$B(P, Q) = \sum_x \sqrt{P(x)Q(x)}$$

or the corresponding integral. The squared classical fidelity is $F(P, Q) = B(P, Q)^2$. For probability distributions we use the total-variation distance

$$\|P - Q\|_{\text{TV}} := \sup_A |P(A) - Q(A)| = \frac{1}{2} \|P - Q\|_1. \quad (5)$$

Thus total variation is the classical trace distance; the factor $1/2$ is a convention that we fix once and use throughout.

We will use the following standard facts.

Lemma 2.1 (Continuity and monotonicity). *Let Λ be a quantum channel. Then*

$$B(\Lambda(\rho), \Lambda(\sigma)) \geq B(\rho, \sigma).$$

Root fidelity is jointly concave. Moreover, if $\|\rho - \rho'\|_1 \leq \varepsilon$ and $\|\sigma - \sigma'\|_1 \leq \varepsilon$, then

$$|B(\rho, \sigma) - B(\rho', \sigma')| \leq C\sqrt{\varepsilon}$$

for a universal constant C .

Proof. Monotonicity and joint concavity are standard consequences of Uhlmann's theorem. The continuity bound follows from the Fuchs–van de Graaf inequalities and the triangle inequality for the Bures distance

$$d_B(\rho, \sigma) = \sqrt{2(1 - B(\rho, \sigma))}.$$

Indeed $d_B(\rho, \rho') \leq \|\rho - \rho'\|_1^{1/2}$, and similarly for σ, σ' . Since $B = 1 - d_B^2/2$, this gives the displayed estimate for a universal constant; only convergence to zero is used in the LAN transfers below. $\square$

If states are block diagonal over a common classical label,

$$\rho = \bigoplus_x P_x \rho_x, \quad \sigma = \bigoplus_x Q_x \sigma_x,$$

then

$$B(\rho, \sigma) = \sum_x \sqrt{P_x Q_x} B(\rho_x, \sigma_x). \quad (6)$$

The following elementary lemma will be used twice: once for the Young labels in the known-spectrum protocol and once for the classical label in the adaptive protocol.

Lemma 2.2 (Combining a classical label with conditional quantum fidelities). *For each n , let $J_n(x, y)$ be a joint probability distribution with y -marginal $Q_n(y)$, let $P_n(y)$ be a target label distribution, and let $\sigma_{n,x,y}$ and $\tau_{n,y}$ be quantum states. Define*

$$\bar{\sigma}_{n,y} = \sum_x \frac{J_n(x, y)}{Q_n(y)} \sigma_{n,x,y}$$

when $Q_n(y) > 0$, with an arbitrary value otherwise. Suppose there are good sets G_n such that

$$J_n(G_n^c) \longrightarrow 0$$

and

$$\inf_{(x,y) \in G_n} B(\sigma_{n,x,y}, \tau_{n,y}) \geq b - o(1)$$

for some $b \in [0, 1]$. Then

$$\liminf_{n \rightarrow \infty} B\left(\bigoplus_y Q_n(y) \bar{\sigma}_{n,y}, \bigoplus_y P_n(y) \tau_{n,y}\right) \geq b \liminf_{n \rightarrow \infty} B(Q_n, P_n). \quad (7)$$

Proof. By (6) and joint concavity of root fidelity,

$$B\left(\bigoplus_y Q_n(y) \bar{\sigma}_{n,y}, \bigoplus_y P_n(y) \tau_{n,y}\right) \geq \sum_y \sqrt{Q_n(y) P_n(y)} \sum_x \frac{J_n(x, y)}{Q_n(y)} B(\sigma_{n,x,y}, \tau_{n,y}).$$

Let

$$r_n(y) = \sum_{x: (x,y) \notin G_n} \frac{J_n(x, y)}{Q_n(y)}$$

be the conditional bad probability. The last display is at least

$$(b - o(1)) \left[B(Q_n, P_n) - \sum_y \sqrt{Q_n(y) P_n(y)} r_n(y) \right].$$

By Cauchy–Schwarz,

$$\sum_y \sqrt{Q_n(y) P_n(y)} r_n(y) \leq \left(\sum_y Q_n(y) r_n(y) \right)^{1/2} \left(\sum_y P_n(y) r_n(y) \right)^{1/2} \leq \sqrt{J_n(G_n^c)}.$$

This tends to zero and proves (7). $\square$

2.2 Werner’s universal pure-state cloner

We recall the pure-state benchmark that underlies much of the comparison with mixed-state constructions. Let the single-system Hilbert space have dimension D , and let

$$P_{\text{sym}}^{(r)} : (\mathbb{C}^D)^{\otimes r} \rightarrow \text{Sym}^r(\mathbb{C}^D)$$

be the projector onto the totally symmetric subspace. Write

$$d_{r,D}^{\text{sym}} := \dim \text{Sym}^r(\mathbb{C}^D) = \binom{r + D - 1}{D - 1}. \quad (8)$$

Werner’s optimal universal pure-state $n \rightarrow m$ cloner is the covariant channel

$$\mathcal{W}_{n \rightarrow m}^{(D)}(X) = \frac{d_{n,D}^{\text{sym}}}{d_{m,D}^{\text{sym}}} P_{\text{sym}}^{(m)}(X \otimes I^{\otimes(m-n)}) P_{\text{sym}}^{(m)}, \quad (9)$$

where X is supported on $\text{Sym}^n(\mathbb{C}^D)$ [Wer98]. The prefactor makes the map trace preserving. For every unknown pure state ψ , its all-site/global fidelity is

$$F_{\text{glob}}^{\text{Werner}}(n, m, D) = \langle \psi |^{\otimes m} \mathcal{W}_{n \rightarrow m}^{(D)}(|\psi\rangle\langle\psi|^{\otimes n}) |\psi\rangle^{\otimes m} = \frac{d_{n,D}^{\text{sym}}}{d_{m,D}^{\text{sym}}} = \frac{\binom{n+D-1}{D-1}}{\binom{m+D-1}{D-1}}. \quad (10)$$

The one-site reduced fidelity of the same universal cloner is

$$F_1^{\text{Werner}}(n, m, D) = \frac{n(m + D) + m - n}{m(n + D)}. \quad (11)$$

For example, $n = 1$, $m = 2$, $D = 2$ gives the familiar $5/6$ single-qubit fidelity. The distinction between (10) and (11) matters: this paper uses the global/all-site figure of merit, so the relevant pure-state comparator is the dimension ratio (10), not the one-site fidelity. In particular, the random-purification construction discussed in Section 6 inherits exactly (10) with $D = dr$.

2.3 Schur–Weyl decomposition and Weyl-sector states

Schur–Weyl duality gives

$$(\mathbb{C}^d)^{\otimes n} \cong \bigoplus_{\lambda \vdash n, \ell(\lambda) \leq d} \mathcal{H}_\lambda \otimes \mathcal{S}_\lambda, \quad (12)$$

where $\mathcal{H}_\lambda$ is the irreducible $U(d)$ -module of highest weight λ , and $\mathcal{S}_\lambda$ is the corresponding Specht module. For a diagonal density matrix $\rho_p = \text{diag}(p_1, \dots, p_d)$,

$$\rho_p^{\otimes n} = \bigoplus_{\lambda \vdash n} P_{n,p}(\lambda) \rho_{p,\lambda} \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda}, \quad (13)$$

where

$$P_{n,p}(\lambda) = \dim(\mathcal{S}_\lambda) s_\lambda(p), \quad \rho_{p,\lambda} = \frac{\pi_\lambda(\rho_p)}{s_\lambda(p)}. \quad (14)$$

Here s_λ is the Schur polynomial and π_λ is the polynomial representation. For an orbit point $\rho_{p,g} = U(g)\rho_p U(g)^*$, the same decomposition holds with

$$\rho_{p,g,\lambda} = U_\lambda(g) \rho_{p,\lambda} U_\lambda(g)^*.$$

The Young distribution $P_{n,p}$ depends on the spectrum p , but not on the eigenbasis g .

Lemma 2.3 (Young concentration). *Fix a compact subset of the nondegenerate full-rank Weyl chamber,*

$$\Delta_{d,\kappa} = \{p : p_d \geq \kappa, p_i - p_{i+1} \geq \kappa, \sum_i p_i = 1\}.$$

There are constants $c, C > 0$, depending only on d and κ , such that for all $p \in \Delta_{d,\kappa}$,

$$P_{n,p} \left(\left\| \frac{\lambda}{n} - p \right\|_1 > \delta \right) \leq C(n+1)^{d(d-1)/2} \exp(-cn\delta^2) \quad (15)$$

for all sufficiently small $\delta > 0$. In particular, for

$$\mathsf{T}_n(p) = \left\{ \lambda : \left\| \frac{\lambda}{n} - p \right\|_\infty \leq n^{-1/3} \right\},$$

one has $P_{n,p}(\mathsf{T}_n(p)^c) \leq \exp(-c'n^{1/3})$.

Proof. This is the Keyl–Werner spectrum-estimation bound applied to the Young-frame measurement [KW01]. The polynomial prefactor and the exponential rate follow from their large-deviation estimate. The final statement follows by setting $\delta = dn^{-1/3}$ and absorbing the polynomial factor into the exponential. $\square$

2.4 The PRV component and the sector channel

We record the representation-theoretic object used throughout the protocol. The acronym PRV stands for Parthasarathy–Ranga Rao–Varadarajan. The full PRV theorem describes extremal irreducible summands in tensor products of highest-weight representations [PRV67]. The only case needed here is the Cartan, or top, component. If α and β are dominant weights for $U(d)$, then

$$\mathcal{H}_{\alpha+\beta} \subset \mathcal{H}_\alpha \otimes \mathcal{H}_\beta \quad (16)$$

with multiplicity one. We call the corresponding normalized intertwining isometry

$$V_{\alpha,\beta} : \mathcal{H}_{\alpha+\beta} \hookrightarrow \mathcal{H}_\alpha \otimes \mathcal{H}_\beta, \quad V_{\alpha,\beta} v_{\alpha+\beta} = v_\alpha \otimes v_\beta, \quad (17)$$

the PRV inclusion. Thus the phrase “PRV channel” in this paper always refers to the Cartan/PRV component; no other PRV summand is used.

If $\mu \vdash n$, $\nu \vdash m$, and $\omega := \nu - \mu$ is dominant, then $\nu = \mu + \omega$ and (17) gives

$$V_{\mu,\omega} : \mathcal{H}_\nu \hookrightarrow \mathcal{H}_\mu \otimes \mathcal{H}_\omega. \quad (18)$$

The associated sector channel, used later in Definition 4.2, is

$$\mathcal{C}_{\mu \rightarrow \nu}^{\text{PRV}}(X) = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} V_{\mu,\omega}^*(X \otimes I_\omega) V_{\mu,\omega}, \quad X \in \mathcal{B}(\mathcal{H}_\mu). \quad (19)$$

This is a channel from $\mathcal{B}(\mathcal{H}_\mu)$ to $\mathcal{B}(\mathcal{H}_\nu)$. Complete positivity is immediate from the Kraus form obtained by inserting an orthonormal basis of $\mathcal{H}_\omega$; trace preservation follows from Schur’s lemma because $\text{Tr}_\omega(V_{\mu,\omega} V_{\mu,\omega}^*)$ is a scalar multiple of the identity on the irreducible space $\mathcal{H}_\mu$, and its trace fixes the scalar to $\dim \mathcal{H}_\nu / \dim \mathcal{H}_\mu$. This early definition is repeated in the protocol section only to keep the construction self-contained.

3 Quantum statistical models, LAN, and Gaussian cloning

This section introduces the limiting statistical models used in the converse proofs. The terminology is standard in quantum statistics, but we give the definitions needed here because the finite-dimensional Schur–Weyl problem and its Gaussian limit live on rather different Hilbert spaces.

3.1 Classical and quantum statistical models

Definition 3.1 (Statistical model). *A classical statistical model is a family of probability distributions*

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}$$

indexed by an unknown parameter θ . A quantum statistical model is a family of density operators

$$\mathcal{Q} = \{\rho_\theta : \theta \in \Theta\}$$

on a common Hilbert space. In both cases the word “model” means the whole parameterized family, not a single distribution or state.

A randomized classical procedure is described by a Markov kernel. If X is an input space and Y an output space, a Markov kernel $K(dy|x)$ assigns to every input x a probability distribution on Y , measurably in x . On finite spaces it is simply a stochastic matrix:

$$K(y|x) \geq 0, \quad \sum_y K(y|x) = 1.$$

Thus the kernel $K_n(\nu|\mu)$ used later is just a conditional probability for the output Young diagram ν given the measured input diagram μ . The quantum analogue is a quantum channel: a completely positive trace-preserving linear map on trace-class operators.

A channel transforms one statistical model into another by acting on every member of the family. Two sequences of models are asymptotically equivalent if there are channels in both directions whose worst-case trace-norm errors tend to zero. Local asymptotic normality is such an equivalence after zooming into a $1/\sqrt{n}$ neighborhood of a fixed state.

3.2 Bosonic modes and displaced thermal states

A single bosonic mode has Fock space

$$\mathcal{F} = \text{span}\{|k\rangle : k = 0, 1, 2, \dots\}.$$

The annihilation and creation operators are defined by

$$a|k\rangle = \sqrt{k}|k-1\rangle, \quad a^*|k\rangle = \sqrt{k+1}|k+1\rangle,$$

and satisfy the canonical commutation relation

$$[a, a^*] = I.$$

The number operator $N = a^*a$ has eigenvalue k on $|k\rangle$. Phase rotations are

$$R(\theta) = e^{i\theta N}.$$

For $z \in \mathbb{C}$, the Weyl displacement operator is

$$D(z) = \exp(za^* - \bar{z}a). \tag{20}$$

It translates the canonical mode:

$$D(z)^* a D(z) = a + z.$$

For $0 < q < 1$, the thermal state is

$$\tau_q = (1-q) \sum_{k \geq 0} q^k |k\rangle\langle k|. \tag{21}$$

Its mean photon number is

$$\bar{n}_q = \text{Tr}(\tau_q N) = \frac{q}{1-q}.$$

A displaced thermal state is $D(z)\tau_q D(z)^*$. These are quantum Gaussian states: their Weyl characteristic function

$$\chi_\rho(w) = \text{Tr}[\rho D(w)]$$

is a Gaussian function of the real and imaginary parts of w . No other class of Gaussian states is needed in this paper.

Because two thermal states are diagonal in the same number basis, their root fidelity is the classical Hellinger affinity of two geometric distributions:

$$B(\tau_q, \tau_{q'}) = \sum_{k \geq 0} \sqrt{(1-q)(1-q')q^k(q')^k} = \frac{\sqrt{(1-q)(1-q')}}{1 - \sqrt{qq'}}. \quad (22)$$

Consequently,

$$F(\tau_q, \tau_{q'}) = \frac{(1-q)(1-q')}{(1 - \sqrt{qq'})^2}. \quad (23)$$

For R independent modes we write

$$\mathcal{F}^{\otimes R}, \quad D(z) = \bigotimes_{\alpha=1}^R D(z_\alpha), \quad z = (z_1, \dots, z_R) \in \mathbb{C}^R.$$

All oscillator modes appearing in the LAN limit below are of this independent product form.

3.3 Phase-insensitive quantum amplification

We next define the Gaussian channel that appears both in the achievable limit of the PRV map and in the converse.

Definition 3.2 (Quantum-limited phase-insensitive amplifier). *Let $G \geq 1$. Introduce an environment mode b , initially in its vacuum state $|0\rangle$, and a two-mode squeezing unitary U_G satisfying*

$$U_G^*(a \otimes I)U_G = \sqrt{G}a \otimes I + \sqrt{G-1}I \otimes b^*. \quad (24)$$

The quantum-limited phase-insensitive amplifier of intensity gain G is

$$\mathcal{A}_G(\rho) = \text{Tr}_{\text{env}} [U_G(\rho \otimes |0\rangle\langle 0|)U_G^*].$$

The coefficient $\sqrt{G-1}$ is forced by preservation of the commutation relation:

$$[\sqrt{G}a + \sqrt{G-1}b^*, \sqrt{G}a^* + \sqrt{G-1}b] = G - (G-1) = 1.$$

The term “quantum limited” means that the environment is in the vacuum, so the channel adds the minimum noise compatible with this commutation relation. The channel is *phase insensitive* because it treats all phase-space directions equally:

$$\mathcal{A}_G(R(\theta)\rho R(\theta)^*) = R(\theta)\mathcal{A}_G(\rho)R(\theta)^*.$$

It is also displacement covariant with amplitude gain $\sqrt{G}$:

$$\mathcal{A}_G(D(z)\rho D(z)^*) = D(\sqrt{G}z)\mathcal{A}_G(\rho)D(\sqrt{G}z)^*. \quad (25)$$

On a thermal input, the output mean photon number is

$$\bar{n}_{\text{out}} = G\bar{n}_q + (G-1).$$

Writing the output again as a thermal state, $q_G = \bar{n}_{\text{out}}/(1 + \bar{n}_{\text{out}})$, gives

$$\mathcal{A}_G(D(z)\tau_q D(z)^*) = D(\sqrt{G}z)\tau_{q_G} D(\sqrt{G}z)^*, \quad q_G = 1 - \frac{1-q}{G}. \quad (26)$$

Later $G = \gamma = m_n/n$. Notice that the amplifier reproduces the correct enlarged displacement $\sqrt{\gamma}z$, but it necessarily changes q to the noisier value

$$q^{(\gamma)} := 1 - \frac{1-q}{\gamma}.$$

A general phase-insensitive amplifier in the converse need not be Gaussian. It means any quantum channel satisfying the displacement covariance (25) and phase covariance above. The Guță–Matsumoto theorem used below says that, on displaced thermal inputs, the quantum-limited amplifier is least noisy among all such channels [GM06].

Definition 3.3 (Gaussian input, target, and least-noise output families). *For one oscillator mode, thermal parameter $q \in (0, 1)$, displacement $u \in \mathbb{C}$, and gain $\gamma > 1$, define*

$$\phi_u^{(q)} := D(u)\tau_q D(u)^*, \quad (27)$$

$$\psi_u^{(q,\gamma)} := D(\sqrt{\gamma}u)\tau_q D(\sqrt{\gamma}u)^*, \quad (28)$$

$$\omega_u^{(q,\gamma)} := D(\sqrt{\gamma}u)\tau_{q^{(\gamma)}} D(\sqrt{\gamma}u)^*, \quad (29)$$

where

$$q^{(\gamma)} = 1 - \frac{1-q}{\gamma}.$$

The state $\psi_u^{(q,\gamma)}$ is the ideal amplified target: it has the desired displacement $\sqrt{\gamma}u$ but the original thermal noise. The state $\omega_u^{(q,\gamma)}$ is the output of the quantum-limited amplifier; it has the correct displacement but necessarily larger thermal noise.

For the fixed-spectrum orbit, let

$$R = \frac{d(d-1)}{2}, \quad z = (z_{ij})_{i < j} \in \mathbb{C}^R, \quad q_{ij} = \frac{p_j}{p_i}.$$

Define the multimode input, ideal target, and least-noise output families by

$$\Phi_z^{(p)} := \bigotimes_{i < j} \phi_{z_{ij}}^{(q_{ij})}, \quad (30)$$

$$\Psi_z^{(p,\gamma)} := \bigotimes_{i < j} \psi_{z_{ij}}^{(q_{ij},\gamma)}, \quad (31)$$

$$\Omega_z^{(p,\gamma)} := \bigotimes_{i < j} \omega_{z_{ij}}^{(q_{ij},\gamma)}. \quad (32)$$

When p and γ are fixed, we suppress the superscripts and write simply Φ_z, Ψ_z, Ω_z .

3.4 Local asymptotic normality

We now explain how bosonic modes arise from many finite-dimensional copies. The results in this subsection are the finite-dimensional quantum LAN theorem of Kahn and Guță, specialized to the parameterizations used here [KG09]. Fix a full-rank state $\rho_p = \text{diag}(p_1, \dots, p_d)$ with strictly ordered eigenvalues. A local perturbation of size $1/\sqrt{n}$ is statistically visible from n copies: smaller perturbations disappear asymptotically, while larger perturbations become perfectly distinguishable. LAN describes the n -copy family on exactly this scale.

We now choose local coordinates for the two kinds of perturbations of ρ_p : changes of eigenvectors and changes of eigenvalues. Let $E_{jk} = |j\rangle\langle k|$, and for $j < k$ define the Hermitian generators

$$T_{jk} = i(E_{jk} - E_{kj}), \quad T_{kj} = E_{jk} + E_{kj}.$$

For

$$z = (z_{jk})_{j < k} \in \mathbb{C}^R, \quad R = \frac{d(d-1)}{2},$$

set

$$X_p(z) = i \sum_{j < k} \frac{\Re z_{jk} T_{jk} + \Im z_{jk} T_{kj}}{\sqrt{p_j - p_k}}. \quad (33)$$

The normalization in $X_p(z)$ deserves explanation. Let

$$\delta_{jk} = p_j - p_k.$$

Then

$$X_p(z) = \sum_{j < k} \frac{z_{jk} E_{kj} - \bar{z}_{jk} E_{jk}}{\sqrt{\delta_{jk}}}.$$

Consequently,

$$[X_p(z), \rho_p]_{kj} = \sqrt{\delta_{jk}} z_{jk}.$$

Thus z_{jk} is not the raw off-diagonal matrix perturbation: the latter is $\sqrt{\delta_{jk}} z_{jk} / \sqrt{n}$. The gap normalization is chosen instead to match the oscillator normalization in the LAN limit.

Indeed, in a typical Schur sector μ , let

$$\Delta_{jk}^\mu = \mu_j - \mu_k, \quad a_{jk}^{(\mu)\dagger} = \frac{E_{kj}^{(\mu)}}{\sqrt{\Delta_{jk}^\mu}}, \quad a_{jk}^{(\mu)} = \frac{E_{jk}^{(\mu)}}{\sqrt{\Delta_{jk}^\mu}}.$$

Then

$$\frac{d\pi_\mu(X_p(z))}{\sqrt{n}} = \sum_{j < k} \sqrt{\frac{\Delta_{jk}^\mu}{n(p_j - p_k)}} \left(z_{jk} a_{jk}^{(\mu)\dagger} - \bar{z}_{jk} a_{jk}^{(\mu)} \right).$$

For typical μ ,

$$\frac{\Delta_{jk}^\mu}{n(p_j - p_k)} \rightarrow 1,$$

so the right-hand side converges on fixed excitation cutoffs to

$$\sum_{j < k} \left(z_{jk} a_{jk}^\dagger - \bar{z}_{jk} a_{jk} \right).$$

Hence

$$U_\mu(\exp(X_p(z)/\sqrt{n})) \longrightarrow \bigotimes_{j < k} D(z_{jk}).$$

This is the precise sense in which z_{jk} is the limiting oscillator displacement amplitude.

Moreover,

$$\frac{X_p(z)}{\sqrt{n}} = \frac{X_p(\sqrt{m_n/n} z)}{\sqrt{m_n}}.$$

Thus in the m_n -copy LAN chart the same physical unitary has coordinate $\sqrt{m_n/n} z \rightarrow \sqrt{\gamma} z$. This gives the amplitude gain $\sqrt{\gamma}$ in the Gaussian cloning target.

The matrix $X_p(z)$ is anti-Hermitian, so $\exp(X_p(z))$ is unitary. The complex coordinate z_{jk} contains the two real infinitesimal rotations that mix the j -th and k -th eigenvectors. The factor $(p_j - p_k)^{-1/2}$ is chosen so that, after normalizing the Schur-sector root operators by $\sqrt{\mu_j - \mu_k}$, the local unitary converges to the Weyl displacement $D(z_{jk})$.

For spectral perturbations, let

$$\mathbf{H} = \left\{ h \in \mathbb{R}^d : \sum_i h_i = 0 \right\}, \quad p_{n,h} = p + \frac{h}{\sqrt{n}}.$$

The condition $\sum_i h_i = 0$ preserves normalization. For every bounded set of h 's, positivity and strict ordering are automatic for all sufficiently large n , because $p_d > 0$ and $\min_i(p_i - p_{i+1}) > 0$. More explicitly, if $\|h\|_\infty \leq L$, then it is enough that

$$\frac{L}{\sqrt{n}} < p_d, \quad \frac{2L}{\sqrt{n}} < \min_i(p_i - p_{i+1}).$$

Thus the phrase “ $p + h/\sqrt{n}$ is a state” simply means that its entries are nonnegative and sum to one; in the compact local regimes used below, this holds automatically once n is large. The covariance matrix of the spectral fluctuations is

$$V_p = \text{diag}(p) - pp^T$$

restricted to $\mathbf{H}$.

The corresponding one-copy local state is

$$\rho_{p,h,z}^{(n)} = g_{n,z} \text{diag}\left(p + \frac{h}{\sqrt{n}}\right) g_{n,z}^*, \quad g_{n,z} = \exp\left(\frac{X_p(z)}{\sqrt{n}}\right). \quad (34)$$

Expanding to first order,

$$\rho_{p,h,z}^{(n)} = \rho_p + \frac{1}{\sqrt{n}} \left(\text{diag}(h) + [X_p(z), \rho_p] \right) + O(n^{-1}).$$

Hence h changes the eigenvalues while z changes the eigenbasis. The scale $n^{-1/2}$ is the local statistical scale on which n copies have a nontrivial Gaussian limit.

Theorem 3.4 (Fixed-spectrum LAN). *Fix $p \in \Delta_d^\circ$, and let $\Phi_z = \Phi_z^{(p)}$ and $\Psi_z = \Psi_z^{(p,\gamma)}$ be the Gaussian input and target families from [Definition 3.3](#). With*

$$g_{n,z} = \exp(X_p(z)/\sqrt{n}), \quad \rho_{n,z} = \rho_{p,g_{n,z}}^{\otimes n},$$

there are quantum channels

$$T_n : (\mathbb{C}^d)^{\otimes n} \longrightarrow \mathcal{F}^{\otimes R}, \quad S_n : \mathcal{F}^{\otimes R} \longrightarrow (\mathbb{C}^d)^{\otimes n},$$

such that, uniformly for z in every fixed compact subset of $\mathbb{C}^R$,

$$\|T_n(\rho_{n,z}) - \Phi_z\|_1 \longrightarrow 0, \quad (35)$$

$$\|S_n(\Phi_z) - \rho_{n,z}\|_1 \longrightarrow 0. \quad (36)$$

If $m_n/n \rightarrow \gamma$, then the same physical group element, acting on m_n copies, satisfies

$$\|T_{m_n}(\rho_{p,g_{n,z}}^{\otimes m_n}) - \Psi_z\|_1 \longrightarrow 0 \quad (37)$$

uniformly on compact z -sets.

The maps T_n and S_n are called the forward and inverse LAN channels. The first converts the finite-dimensional experiment into its oscillator approximation; the second reconstructs finite-dimensional states from the Gaussian model. The two trace-norm limits mean that, for bounded local parameters, neither model contains asymptotically more statistical information than the other.

The factor $\sqrt{\gamma}$ in (37) is simply a change of local scale. The group element is written as $\exp(X_p(z)/\sqrt{n})$. In an m_n -copy chart normalized by $1/\sqrt{m_n}$, the same group element has coordinate $\sqrt{m_n/n}z \rightarrow \sqrt{\gamma}z$. The thermal parameter remains q_{ij} , since it depends only on p .

Theorem 3.5 (Full mixed-state LAN). *Fix $p \in \Delta_d^\circ$. For*

$$p_{n,h} = p + h/\sqrt{n}, \quad g_{n,z} = \exp(X_p(z)/\sqrt{n}),$$

the n -copy quantum statistical model converges, uniformly for (h, z) in compact subsets of $\mathbf{H} \times \mathbb{C}^R$, to the hybrid classical–quantum model

$$\mathbf{N}_{h,V_p} \otimes \Phi_z. \quad (38)$$

Here $\mathbf{N}_{h,V_p}$ is the ordinary Gaussian probability distribution on $\mathbf{H}$ with mean h and covariance V_p . The product symbol means that this classical variable and the oscillator state are independent components of one hybrid classical–quantum statistical model. The m_n -copy target converges to

$$\mathbf{N}_{\sqrt{\gamma}h,V_p} \otimes \Psi_z. \quad (39)$$

There are forward and inverse channels whose worst-case trace-norm errors tend to zero on every compact local parameter set.

Thus the full LAN model separates into two independent pieces. The classical Gaussian variable records $1/\sqrt{n}$ -scale changes of the eigenvalues. The R oscillator modes record the noncommuting changes of eigenvectors. The fixed-spectrum theorem is the restriction obtained by freezing $h = 0$.

Corollary 3.6 (Compact-uniform LAN on strict chambers). *Fix $\kappa > 0$ and $L < \infty$. The convergences in Theorems 3.4 and 3.5 are uniform over*

$$p \in \Delta_{d,\kappa} = \left\{ p : p_d \geq \kappa, p_i - p_{i+1} \geq \kappa, \sum_i p_i = 1 \right\}$$

and $(h, z) \in C_L$. For all sufficiently large n , every such $p + h/\sqrt{n}$ is automatically a full-rank, strictly ordered probability vector.

Proof. Kahn–Guță’s construction and error bounds are local-uniform on compact subsets of the full-rank, nondegenerate state space [KG09]. On $\Delta_{d,\kappa}$, all eigenvalues and all spectral gaps are bounded away from zero. The coordinate charts, the classical covariance V_p , and the oscillator temperatures $q_{ij} = p_j/p_i$ therefore range over compact sets. Cover $\Delta_{d,\kappa}$ by finitely many LAN neighborhoods and take the maximum of the corresponding error functions. $\square$

3.5 Transferring a cloning bound through LAN

The converse proofs use a simple but important stability statement. It is stated separately so that every use of LAN later reduces to an explicit inequality.

Lemma 3.7 (LAN fidelity transfer). *Let $\{\rho_{n,\theta}\}_{\theta \in K}$ and $\{\sigma_{n,\theta}\}_{\theta \in K}$ be finite-dimensional input and target models on a compact parameter set K . Let $\{\Phi_\theta\}$ and $\{\Psi_\theta\}$ be limiting models, and suppose channels S_n, T_n satisfy*

$$\sup_{\theta \in K} \|S_n(\Phi_\theta) - \rho_{n,\theta}\|_1 \rightarrow 0, \quad \sup_{\theta \in K} \|T_n(\sigma_{n,\theta}) - \Psi_\theta\|_1 \rightarrow 0.$$

For any sequence of finite-dimensional channels $\mathcal{M}_n$, define

$$\Gamma_n = T_n \circ \mathcal{M}_n \circ S_n.$$

Then

$$\inf_{\theta \in K} B(\Gamma_n(\Phi_\theta), \Psi_\theta) \geq \inf_{\theta \in K} B(\mathcal{M}_n(\rho_{n,\theta}), \sigma_{n,\theta}) - o(1). \quad (40)$$

Proof. Let

$$\delta_n = \sup_{\theta \in K} \|S_n(\Phi_\theta) - \rho_{n,\theta}\|_1, \quad \eta_n = \sup_{\theta \in K} \|T_n(\sigma_{n,\theta}) - \Psi_\theta\|_1.$$

Trace-norm contractivity gives

$$\|\Gamma_n(\Phi_\theta) - T_n \mathcal{M}_n(\rho_{n,\theta})\|_1 \leq \delta_n.$$

By continuity of root fidelity from [Lemma 2.1](#),

$$B(\Gamma_n(\Phi_\theta), \Psi_\theta) \geq B(T_n \mathcal{M}_n(\rho_{n,\theta}), T_n(\sigma_{n,\theta})) - C(\sqrt{\delta_n} + \sqrt{\eta_n}).$$

Monotonicity under the channel T_n gives

$$B(T_n \mathcal{M}_n(\rho_{n,\theta}), T_n(\sigma_{n,\theta})) \geq B(\mathcal{M}_n(\rho_{n,\theta}), \sigma_{n,\theta}).$$

Taking the infimum over $\theta \in K$ proves (40). $\square$

3.6 The displaced-thermal Gaussian cloning problem

The finite-dimensional converse will now follow from a cloning theorem for the limit model. We use the notation introduced in [Definition 3.3](#). For one mode, $\phi_u^{(q)}$ is the displaced thermal input, $\psi_u^{(q,\gamma)}$ is the ideal target with amplified displacement, and $\omega_u^{(q,\gamma)}$ is the output of the quantum-limited amplifier. For the fixed-spectrum multimode model, the corresponding product families are Φ_z, Ψ_z, Ω_z .

A probability distribution P on $\mathbb{N}$ is said to be stochastically larger than P^* if

$$\sum_k \phi(k) P_k \geq \sum_k \phi(k) P_k^*$$

for every bounded increasing function ϕ . Equivalently, expectations of bounded decreasing functions are no larger under P . Guță and Matsumoto show that, among phase-insensitive displacement-covariant amplifiers of fixed gain, the quantum-limited output photon-number distribution is smallest in this stochastic order [\[GM06\]](#).

Lemma 3.8 (Uniform trace reduction for covariant CP maps). *Let $\mathcal{J}$ be a normal completely positive trace-nonincreasing map on an oscillator trace class and suppose*

$$\mathcal{J}(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\mathcal{J}(X)D(\sqrt{\gamma}z)^*.$$

Then there is $c \in [0, 1]$ such that

$$\text{Tr } \mathcal{J}(X) = c \text{Tr } X$$

for every trace-class X .

Proof. The output trace is a positive normal linear functional, so

$$\mathrm{Tr} \mathcal{J}(X) = \mathrm{Tr}[AX]$$

for a bounded operator $0 \leq A \leq I$. Covariance and cyclicity imply $D(z)^*AD(z) = A$ for all z . The Weyl representation on Fock space is irreducible, so its commutant is the scalar algebra. Hence $A = cI$. $\square$

Lemma 3.9 (Hunt–Stein reduction for the Gaussian models). *For the flat-prior cloning problems considered below, the asymptotic value over expanding cubes is unchanged if one restricts the optimization to exactly covariant channels.*

Proof. The relevant covariance groups are $\mathbb{C}^R$ and $\mathbb{H} \times \mathbb{C}^R$. They are locally compact abelian groups and hence amenable. The Hunt–Stein theorem for amenable covariant decision problems applies because the loss is invariant and root fidelity is concave in the output state; see [Hol11]. Equivalently, one may average a channel over a Følner sequence of cubes. Amenability guarantees an exactly covariant decision rule with value at least the limiting flat-prior value, avoiding normalization issues in informal weak-limit arguments. $\square$

Lemma 3.10 (One-mode conic stochastic order). *Let $\mathcal{J}$ be a normal completely positive trace-nonincreasing one-mode map that is displacement covariant with gain $\sqrt{\gamma}$ and phase covariant. Then, for every bounded nonnegative decreasing function $w : \mathbb{N} \rightarrow \mathbb{R}_+$,*

$$\mathrm{Tr}[\mathcal{J}(\tau_q)w(N)] \leq \mathrm{Tr}[\mathcal{J}(\tau_q)] \mathrm{Tr}[\tau_{q(\gamma)}w(N)]. \quad (41)$$

Proof. By Lemma 3.8,

$$\mathrm{Tr} \mathcal{J}(X) = c \mathrm{Tr} X$$

for some $c \in [0, 1]$. If $c = 0$, the map is zero. If $c > 0$, then $c^{-1}\mathcal{J}$ is a trace-preserving phase-insensitive displacement-covariant amplifier. Guță and Matsumoto characterize displacement-covariant amplifiers by an idler state; phase covariance reduces the idler to a number-diagonal state. Their stochastic-order definition and least-noise lemma [GM06, Definition 2.2 and Lemma 2.3], together with the arbitrary-gain extension stated in [GM06, Sec. III], imply that the output photon-number law of $c^{-1}\mathcal{J}(\tau_q)$ stochastically dominates the geometric law of $\tau_{q(\gamma)}$. Equivalently, the expectation of every bounded decreasing photon-number function is no larger than for the quantum-limited output. Multiplying by c proves (41). $\square$

Remark 3.11 (The one external Gaussian input). *The only stochastic-order statement imported from Guță–Matsumoto is their one-mode trace-preserving result, namely Definition 2.2 and Lemma 2.3 together with the arbitrary-gain extension in their Section III [GM06]. Their Theorem 2.5 uses that order to optimize trace distance; our Uhlmann-fidelity bound is derived separately in Lemma 3.12. The multimode estimate below is proved internally by a peeling argument, so no independent multivariate stochastic-order theorem is assumed.*

Lemma 3.12 (Fidelity optimality in one mode). *Among displacement-covariant amplifiers of gain γ , the quantum-limited amplifier maximizes root fidelity with the one-mode target family $\psi_u^{(q,\gamma)}$. The optimal squared fidelity is*

$$f_\gamma(q) = \frac{(1-q)(1-q^{(\gamma)})}{(1-\sqrt{qq^{(\gamma)}})^2}. \quad (42)$$

Proof. Covariance makes the fidelity independent of u , so set $u = 0$. Average the channel over phase rotations. Because the input and target are phase invariant and root fidelity is concave, this cannot lower the fidelity. The output is therefore diagonal:

$$\sigma = \sum_k P_k |k\rangle\langle k|.$$

Let

$$T_k = (1 - q)q^k, \quad P_k^* = (1 - q^{(\gamma)})(q^{(\gamma)})^k.$$

For every $x, x_0 > 0$,

$$\sqrt{x} \leq \sqrt{x_0} + \frac{x - x_0}{2\sqrt{x_0}}.$$

Applying this pointwise to $x = P_k T_k$ and $x_0 = P_k^* T_k$, and summing over k , gives

$$B(P, T) \leq B(P^*, T) + \frac{1}{2} \sum_k \sqrt{\frac{T_k}{P_k^*}} (P_k - P_k^*).$$

The weight

$$w_k = \sqrt{\frac{T_k}{P_k^*}} = \sqrt{\frac{1 - q}{1 - q^{(\gamma)}}} \left(\frac{q}{q^{(\gamma)}} \right)^{k/2}$$

is bounded and decreasing. The one-mode conic order makes the last sum nonpositive. Hence $B(P, T) \leq B(P^*, T)$, and (23) gives (42). $\square$

The equivalent form

$$f_\gamma(q) = \left(\frac{\sqrt{\gamma} + \sqrt{q(\gamma - 1 + q)}}{\gamma + q} \right)^2 \quad (43)$$

follows from $q^{(\gamma)} = (\gamma - 1 + q)/\gamma$.

Lemma 3.13 (Tensorization of the one-mode conic order). *Let $\mathcal{A}$ be an R -mode normal completely positive trace-nonincreasing map, displacement covariant with gain $\sqrt{\gamma}$ and phase covariant in each mode. Let*

$$\Phi_0 = \bigotimes_{\alpha=1}^R \tau_{q_\alpha}, \quad \Omega_{*,Q} = \bigotimes_{\alpha=1}^R \tau_{q_\alpha^{(\gamma)}}.$$

For every product weight

$$W(\mathbf{N}) = \prod_{\alpha=1}^R w_\alpha(N_\alpha)$$

with each w_α bounded, nonnegative, and decreasing,

$$\mathrm{Tr}[\mathcal{A}(\Phi_0)W(\mathbf{N})] \leq \mathrm{Tr}[\mathcal{A}(\Phi_0)] \mathrm{Tr}[\Omega_{*,Q}W(\mathbf{N})]. \quad (44)$$

Proof. By rescaling, assume $0 \leq w_\alpha \leq 1$. Put

$$W_{>1} = \prod_{\alpha=2}^R w_\alpha(N_\alpha), \quad \Phi_{0,>1} = \bigotimes_{\alpha=2}^R \tau_{q_\alpha}.$$

Define the one-mode CP map

$$\mathcal{J}_1(X) = \mathrm{Tr}_{2,\dots,R} \left[(I \otimes W_{>1}^{1/2}) \mathcal{A}(X \otimes \Phi_{0,>1}) (I \otimes W_{>1}^{1/2}) \right].$$

Since $0 \leq W_{>1} \leq I$, this map is trace nonincreasing. The covariance of $\mathcal{A}$ and phase invariance of $W_{>1}$ imply that $\mathcal{J}_1$ has the one-mode covariance required in [Lemma 3.10](#). Therefore

$$\text{Tr}[\mathcal{A}(\Phi_0)w_1(N_1)W_{>1}] \leq \text{Tr}[\tau_{q_1^{(\gamma)}}w_1(N_1)] \text{Tr}[\mathcal{A}(\Phi_0)W_{>1}].$$

Repeat this argument for modes $2, \dots, R$. The resulting product is exactly (44). $\square$

For the fixed-spectrum LAN model define

$$B_G(\gamma, p) := \prod_{i < j} \frac{\sqrt{(1 - q_{ij})(1 - q_{ij}^{(\gamma)})}}{1 - \sqrt{q_{ij}q_{ij}^{(\gamma)}}}, \quad (45)$$

$$F_G(\gamma, p) := B_G(\gamma, p)^2. \quad (46)$$

These equal $B_{\text{orb}}(\gamma, p)$ and $F_{\text{orb}}(\gamma, p)$.

Theorem 3.14 (Gaussian fidelity converse). *Let $\Phi_z = \Phi_z^{(p)}$ and $\Psi_z = \Psi_z^{(p, \gamma)}$ be the multimode Gaussian input and ideal target families from [Definition 3.3](#). Let $C_L \subset \mathbb{C}^R$ be the cube*

$$|\Re z_\alpha|, |\Im z_\alpha| \leq L.$$

Then

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \int_{C_L} B(\Lambda(\Phi_z), \Psi_z) \frac{dz}{|C_L|} = B_G(\gamma, p). \quad (47)$$

Consequently no channel can have minimax squared fidelity larger than $F_G(\gamma, p)$.

Proof. The product quantum-limited amplifier attains the lower bound, with constant root fidelity $B_G(\gamma, p)$. For the upper bound, [Lemma 3.9](#) permits restriction to exactly displacement-covariant channels. Covariance reduces the fidelity to $z = 0$. Phase averaging may again be applied, so the origin output is diagonal in the joint number basis:

$$P_{\mathbf{k}} = \langle \mathbf{k} | \Lambda(\Phi_0) | \mathbf{k} \rangle.$$

Let

$$T_{\mathbf{k}} = \prod_{\alpha} (1 - q_{\alpha}) q_{\alpha}^{k_{\alpha}}, \quad P_{\mathbf{k}}^* = \prod_{\alpha} (1 - q_{\alpha}^{(\gamma)}) (q_{\alpha}^{(\gamma)})^{k_{\alpha}}.$$

Applying the scalar tangent inequality from the proof of [Lemma 3.12](#) to every $\mathbf{k}$ gives

$$B(P, T) \leq B(P^*, T) + \frac{1}{2} \sum_{\mathbf{k}} W_{\mathbf{k}} (P_{\mathbf{k}} - P_{\mathbf{k}}^*),$$

where

$$W_{\mathbf{k}} = C \prod_{\alpha} \left(\frac{q_{\alpha}}{q_{\alpha}^{(\gamma)}} \right)^{k_{\alpha}/2}$$

is a bounded product of decreasing weights. By [Lemma 3.13](#), the second term is nonpositive. Hence

$$B(P, T) \leq B(P^*, T) = B_G(\gamma, p).$$

The minimax bound follows because the infimum of a function over z is no larger than its average over C_L . $\square$

4 Known-spectrum cloning

In this section the spectrum p is fixed and known. Only the orbit element g is unknown. The argument proceeds in four steps. First, we define the global Schur–PRV cloner and prove that its classical Young-label kernel reproduces the target Young distribution asymptotically. Second, we analyze large highest-weight sectors and prove that the PRV map contracts to the beam-splitter dilation of the least-noise Gaussian amplifier. Third, we combine the classical label and conditional sector fidelities to prove achievability. Finally, we use LAN and the Gaussian fidelity converse from [Theorem 3.14](#) to prove the global upper bound for arbitrary cloning channels.

4.1 The known-spectrum Schur–PRV channel

Let $m = m_n$, with $m_n/n \rightarrow \gamma > 1$. Once the input Young diagram μ has been measured, the protocol must choose an output diagram $\nu \vdash m$. This is a purely classical randomized step, hence it is described by a Markov kernel $K_n(\nu|\mu)$ in the sense of [Definition 3.1](#): for each fixed μ , the numbers $K_n(\nu|\mu)$ form a probability distribution over ν .

The PRV sector channel is available only when

$$\omega = \nu - \mu$$

is again a dominant weight. For a typical input diagram, define the admissible set

$$\mathbf{A}_n(\mu) = \{\nu \vdash m : \nu \in \mathbf{T}_m(p), \nu - \mu \text{ is dominant}\}. \quad (48)$$

For $\mu \in \mathbf{T}_n(p)$, set

$$K_n(\nu|\mu) = \frac{P_{m,p}(\nu)}{P_{m,p}(\mathbf{A}_n(\mu))} \mathbf{1}_{\{\nu \in \mathbf{A}_n(\mu)\}}. \quad (49)$$

Thus $K_n(\cdot|\mu)$ is the ideal target Young law conditioned on the event that the sector pair is admissible. On atypical input diagrams we use any fixed fallback output; its choice is asymptotically irrelevant.

Lemma 4.1 (Admissibility and simulation of the target Young law). *For all sufficiently large n ,*

$$\mathbf{T}_m(p) \subseteq \mathbf{A}_n(\mu)$$

for every $\mu \in \mathbf{T}_n(p)$. Consequently,

$$\sup_{\mu \in \mathbf{T}_n(p)} \|K_n(\cdot|\mu) - P_{m,p}\|_{\text{TV}} \longrightarrow 0, \quad (50)$$

$$\|Q_m - P_{m,p}\|_{\text{TV}} \longrightarrow 0, \quad (51)$$

where

$$Q_m(\nu) = \sum_{\mu} P_{n,p}(\mu) K_n(\nu|\mu)$$

is the output Young-label law. Moreover, under the joint law

$$J_n(\mu, \nu) = P_{n,p}(\mu) K_n(\nu|\mu),$$

the complement of the good-pair set

$$G_n = \mathbf{T}_n(p) \times \mathbf{T}_m(p)$$

has probability $J_n(G_n^c) = o(1)$.

Proof. Write

$$\delta_i = p_i - p_{i+1} > 0.$$

If $\mu \in \mathbb{T}_n(p)$, then

$$\mu_i - \mu_{i+1} \leq n\delta_i + 2n^{2/3}.$$

Likewise, if $\nu \in \mathbb{T}_m(p)$, then

$$\nu_i - \nu_{i+1} \geq m\delta_i - 2m^{2/3}.$$

Therefore

$$(\nu_i - \mu_i) - (\nu_{i+1} - \mu_{i+1}) \geq (m - n)\delta_i - 2m^{2/3} - 2n^{2/3}.$$

Since $m/n \rightarrow \gamma > 1$, the first term is of order n , whereas the error is $O(n^{2/3})$. The right-hand side is positive for every $i < d$ and all sufficiently large n . Hence $\nu - \mu$ is dominant and $\mathbb{T}_m(p) \subseteq \mathbb{A}_n(\mu)$.

For a probability distribution P conditioned on an event A ,

$$\|P(\cdot|A) - P\|_{\text{TV}} = P(A^c).$$

Thus, uniformly for typical μ ,

$$\|K_n(\cdot|\mu) - P_{m,p}\|_{\text{TV}} = P_{m,p}(\mathbb{A}_n(\mu)^c) \leq P_{m,p}(\mathbb{T}_m(p)^c) = o(1)$$

by [Lemma 2.3](#). Averaging over μ , and adding the atypical input mass $P_{n,p}(\mathbb{T}_n(p)^c) = o(1)$, proves [\(51\)](#). The same two exceptional events give

$$J_n(G_n^c) \leq P_{n,p}(\mathbb{T}_n(p)^c) + \sup_{\mu \in \mathbb{T}_n(p)} K_n(\mathbb{T}_m(p)^c|\mu) = o(1).$$

□

For $\nu \in \mathbb{A}(\mu)$, set $\omega = \nu - \mu$. As recalled in [Section 2.4](#), the tensor product $\mathcal{H}_\mu \otimes \mathcal{H}_\omega$ contains the Cartan/PRV component of highest weight $\mu + \omega = \nu$. Let

$$V_{\mu,\omega} : \mathcal{H}_\nu \hookrightarrow \mathcal{H}_\mu \otimes \mathcal{H}_\omega \tag{52}$$

be the normalized intertwining isometry satisfying $V_{\mu,\omega}v_\nu = v_\mu \otimes v_\omega$ on highest-weight vectors. We call this the PRV inclusion, following the extremal-weight perspective on Cartan/PRV components [\[PRV67, FH91\]](#).

Definition 4.2 (Sector PRV channel). *The PRV channel from $\mathcal{H}_\mu$ to $\mathcal{H}_\nu$ is*

$$\mathcal{C}_{\mu \rightarrow \nu}^{\text{PRV}}(X) = \frac{d_\mu}{d_\nu} V_{\mu,\omega}^*(X \otimes I_\omega) V_{\mu,\omega}, \quad d_\lambda = \dim \mathcal{H}_\lambda. \tag{53}$$

This formula is in the Schrodinger picture: X is an operator on $\mathcal{H}_\mu$, and the output is an operator on $\mathcal{H}_\nu$. To see complete positivity, choose an orthonormal basis $\{e_a\}$ of $\mathcal{H}_\omega$ and set

$$W_a = (I \otimes \langle e_a |) V_{\mu,\omega} : \mathcal{H}_\nu \rightarrow \mathcal{H}_\mu.$$

Then

$$\mathcal{C}_{\mu \rightarrow \nu}^{\text{PRV}}(X) = \sum_a K_a X K_a^*, \quad K_a = \sqrt{d_\mu/d_\nu} W_a^*.$$

Trace preservation follows from Schur's lemma: $\text{Tr}_\omega(VV^*)$ commutes with the irreducible $\text{U}(d)$ -action on $\mathcal{H}_\mu$, hence equals $(d_\nu/d_\mu)I_\mu$, because its trace is d_ν .

Definition 4.3 (Known-spectrum Schur–PRV cloner). *Given an input state on $(\mathbb{C}^d)^{\otimes n}$:*

- (i) *apply the Schur transform (12);*
- (ii) *measure $\mu \vdash n$;*
- (iii) *discard the input Specht register and keep the Weyl-sector marginal on $\mathcal{H}_\mu$; for the orbit inputs this is lossless, since the Specht factor in (13) is maximally mixed and carries no orbit information;*
- (iv) *sample ν according to (49);*
- (v) *apply $\mathcal{C}_{\mu \rightarrow \nu}^{\text{PRV}}$;*
- (vi) *prepare $I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu$;*
- (vii) *apply the inverse Schur transform on m systems.*

The resulting channel is denoted $\mathcal{C}_{n \rightarrow m}^{\text{PRV},p}$.

For the orbit input, the output in Schur form is

$$\mathcal{C}_{n \rightarrow m}^{\text{PRV},p}(\rho_{p,g}^{\otimes n}) = \bigoplus_{\nu \vdash m} \left[\sum_{\mu \vdash n} P_{n,p}(\mu) K_n(\nu|\mu) \mathcal{C}_{\mu \rightarrow \nu}^{\text{PRV}}(\rho_{p,g,\mu}) \right] \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}. \quad (54)$$

For completeness, here is the same channel on an arbitrary input operator R written directly in Schur form. Let Π_μ be the projection onto $\mathcal{H}_\mu \otimes \mathcal{S}_\mu$, and define the unnormalized Weyl-sector marginal

$$R_\mu := \text{Tr}_{\mathcal{S}_\mu}(\Pi_\mu R \Pi_\mu) \in \mathcal{B}(\mathcal{H}_\mu). \quad (55)$$

Then, up to the harmless fallback choices on atypical or inadmissible sectors,

$$\mathcal{C}_{n \rightarrow m}^{\text{PRV},p}(R) = \bigoplus_{\nu \vdash m} \left[\sum_{\mu \vdash n} K_n(\nu|\mu) \mathcal{C}_{\mu \rightarrow \nu}^{\text{PRV}}(R_\mu) \right] \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}, \quad (56)$$

followed by the inverse Schur transform. Formula (54) is the specialization of (56) to $R = \rho_{p,g}^{\otimes n}$, for which $R_\mu = P_{n,p}(\mu) \rho_{p,g,\mu}$.

4.2 From large highest weights to oscillators

This subsection proves the representation-theoretic input used by the Schur–PRV protocol. The basic idea is simple. A typical Young diagram has row gaps of order n . Excitations obtained by moving only $O(1)$ boxes away from its highest weight then behave like bosonic excitations after the root operators are divided by the square roots of those gaps. The PRV inclusion becomes a beam splitter between these limiting modes.

We spell out the construction because three pieces of terminology are otherwise easy to encounter too abruptly: root operators, PBW monomials, and the Fock-space contraction.

Root excitations and PBW coordinates

For $U(d)$, the positive roots are indexed by pairs

$$R_+ = \{(i, j) : 1 \leq i < j \leq d\}.$$

For $\alpha = (i, j)$, define the lowering and raising operators

$$F_\alpha = E_{ji}, \quad E_\alpha = E_{ij},$$

and the corresponding highest-weight gap

$$\Delta_\alpha^\lambda = \lambda_i - \lambda_j.$$

Acting with F_{ij} lowers the i -th weight and raises the j -th weight by one. Thus it creates one excitation in the root direction (i, j) .

The Poincaré–Birkhoff–Witt (PBW) theorem says that, after choosing an order on the positive roots, ordered monomials in the lowering operators form a basis of the universal enveloping algebra of the lower-triangular Lie algebra [FH91]. Applied to a highest-weight vector v_λ , these monomials span the irreducible representation $\mathcal{H}_\lambda$. We fix a convex root order and write

$$F^{\mathbf{r}} = \prod_{\alpha \in R_+}^{\prec} F_\alpha^{r_\alpha}, \quad |\mathbf{r}| = \sum_\alpha r_\alpha.$$

The multi-index $\mathbf{r}$ records the occupation number of each root mode. For fixed L , the low-excitation subspace is

$$\mathcal{H}_{\lambda, \leq L} = \text{span}\{F^{\mathbf{r}} v_\lambda : |\mathbf{r}| \leq L\}.$$

Only such fixed cutoffs are needed, because the limiting thermal states have exponentially decaying occupation tails.

Lemma 4.4 (Heisenberg contraction of gap-rescaled root operators). *Let $\lambda = \lambda^{(N)}$ range over dominant weights in a compact subcone of the strict Weyl chamber:*

$$cN \leq \Delta_\alpha^\lambda \leq CN \quad (\alpha \in R_+)$$

for constants $0 < c < C < \infty$. Define

$$\hat{F}_\alpha = \frac{F_\alpha}{\sqrt{\Delta_\alpha^\lambda}}, \quad \hat{E}_\alpha = \frac{E_\alpha}{\sqrt{\Delta_\alpha^\lambda}}.$$

On every fixed cutoff $\mathcal{H}_{\lambda, \leq L}$,

$$[\hat{F}_\alpha, \hat{F}_\beta] = O_L(N^{-1/2}), \quad [\hat{E}_\alpha, \hat{E}_\beta] = O_L(N^{-1/2}), \quad (57)$$

$$[\hat{E}_\alpha, \hat{F}_\beta] = \delta_{\alpha\beta} I + O_L(N^{-1/2}) \quad (58)$$

in operator norm, uniformly over λ . Hence the rescaled root operators converge on fixed cutoffs to independent bosonic creation and annihilation operators.

Proof. Each root α generates an $\mathfrak{sl}_2$ subalgebra. On a weight vector lying at most L lowering steps below v_λ , the usual $\mathfrak{sl}_2$ string formula shows

$$\|F_\alpha \psi\| \leq C_L \sqrt{\Delta_\alpha^\lambda} \|\psi\|, \quad \|E_\alpha \psi\| \leq C_L \sqrt{\Delta_\alpha^\lambda} \|\psi\|.$$

Thus $\hat{F}_\alpha$ and $\hat{E}_\alpha$ are uniformly bounded on $\mathcal{H}_{\lambda, \leq L}$.

In type A,

$$[F_\alpha, F_\beta] = N_{\alpha\beta} F_{\alpha+\beta}, \quad N_{\alpha\beta} \in \{0, \pm 1\},$$

with the bracket zero unless $\alpha + \beta$ is a root. When it is a root,

$$[\hat{F}_\alpha, \hat{F}_\beta] = N_{\alpha\beta} \frac{\sqrt{\Delta_{\alpha+\beta}^\lambda}}{\sqrt{\Delta_\alpha^\lambda \Delta_\beta^\lambda}} \hat{F}_{\alpha+\beta}.$$

All gaps are of order N , while $\hat{F}_{\alpha+\beta}$ is uniformly bounded on the cutoff, so the norm is $O_L(N^{-1/2})$. The estimate for two raising operators is the adjoint statement.

For $\alpha = \beta$,

$$[\hat{E}_\alpha, \hat{F}_\alpha] = \frac{H_\alpha}{\Delta_\alpha^\lambda}.$$

On a vector of weight $\lambda - \sum_\beta r_\beta \beta$ with $|\mathbf{r}| \leq L$, the Cartan element H_α acts by

$$\Delta_\alpha^\lambda - \sum_\beta r_\beta \langle \beta, \alpha^\vee \rangle = \Delta_\alpha^\lambda + O_L(1).$$

Hence the normalized commutator is $I + O_L(N^{-1})$. If $\alpha \neq \beta$, then $[E_\alpha, F_\beta]$ is either zero or a single root operator; the same gap normalization gives $O_L(N^{-1/2})$. This proves the lemma. $\square$

Remark 4.5 (Same-weight PBW monomials). *Different PBW monomials can have the same total weight. In $\mathfrak{gl}_3$, for example, $F_{13}v_\lambda$ and $F_{12}F_{23}v_\lambda$ have the same weight. After normalization their overlap is*

$$-\sqrt{\frac{\Delta_{23}}{\Delta_{12}\Delta_{13}}} = O(N^{-1/2})$$

up to the sign convention. Thus the contraction also separates same-weight monomials.

Lemma 4.6 (Uniform PBW/Fock normal form). *Under the assumptions of Lemma 4.4, define*

$$\xi_{\mathbf{r}}^\lambda = \prod_{\alpha \in R_+}^< \frac{F_\alpha^{r_\alpha}}{\sqrt{r_\alpha!} (\Delta_\alpha^\lambda)^{r_\alpha/2}} v_\lambda, \quad |\mathbf{r}| \leq L. \quad (59)$$

Then

$$\langle \xi_{\mathbf{r}}^\lambda, \xi_{\mathbf{s}}^\lambda \rangle = \delta_{\mathbf{r}, \mathbf{s}} + O_L(N^{-1/2}) \quad (60)$$

uniformly on the cutoff. After orthonormalizing separately in each weight space, one obtains vectors $e_{\mathbf{r}}^\lambda$ such that

$$\hat{F}_\alpha e_{\mathbf{r}}^\lambda = \sqrt{r_\alpha + 1} e_{\mathbf{r} + \mathbf{e}_\alpha}^\lambda + O_L(N^{-1/2}), \quad (61)$$

$$\hat{E}_\alpha e_{\mathbf{r}}^\lambda = \sqrt{r_\alpha} e_{\mathbf{r} - \mathbf{e}_\alpha}^\lambda + O_L(N^{-1/2}). \quad (62)$$

Thus $\mathcal{H}_{\lambda, \leq L}$ converges to the L -excitation cutoff of a product of $|R_+|$ Fock spaces.

Proof. Consider the Shapovalov matrix element

$$\langle F^{\mathbf{r}} v_\lambda, F^{\mathbf{s}} v_\lambda \rangle = \langle v_\lambda, (F^{\mathbf{r}})^* F^{\mathbf{s}} v_\lambda \rangle.$$

Commute every raising operator in $(F^{\mathbf{r}})^*$ to the right. Only terms containing Cartan operators survive on v_λ . The term of highest total degree in the gaps is obtained by pairing each E_α directly with a matching F_α . If $\mathbf{r} = \mathbf{s}$, repeated use of

$$[E_\alpha, F_\alpha^r] = r F_\alpha^{r-1} (H_\alpha - r + 1)$$

gives the leading term

$$\prod_{\alpha} r_{\alpha}! (\Delta_{\alpha}^{\lambda})^{r_{\alpha}}.$$

If $\mathbf{r} \neq \mathbf{s}$, at least one nonmatching commutator is required, even when the two monomials have the same total weight. Such a commutator replaces two root operators by one root operator or produces a composite root. Therefore it lowers the total degree in the gaps by at least one. Since every gap is comparable to N , division by the normalization in (59) gives the uniform error $O_L(N^{-1/2})$; the weaker half-power is sufficient for all later arguments.

The cutoff Gram matrix is therefore $G_\lambda = I + O_L(N^{-1/2})$. Because its dimension depends only on L and d , functional calculus gives

$$G_\lambda^{-1/2} = I + O_L(N^{-1/2}).$$

Applying $G_\lambda^{-1/2}$ within each weight space produces the orthonormal basis $e_{\mathbf{r}}^{\lambda}$. Multiplication by F_α , followed by the same commutator calculation, has leading coefficient $\sqrt{r_\alpha + 1}$; its adjoint has leading coefficient $\sqrt{r_\alpha}$. This proves (61) and (62). $\square$

Thermal Weyl-sector states

Lemma 4.7 (Thermal limit of a Weyl-sector Gibbs state). *Let $|\lambda_N| = N$ and $\lambda_N/N \rightarrow p$, where $p_1 > \dots > p_d > 0$. Under the PBW/Fock identification above,*

$$\rho_{p, \lambda_N} \longrightarrow \bigotimes_{\alpha \in R_+} \tau_{q_\alpha}, \quad q_{ij} = \frac{p_j}{p_i}, \quad (63)$$

in trace norm.

Proof. The vector $F^{\mathbf{r}} v_\lambda$ has weight

$$\lambda - \sum_{\alpha} r_{\alpha} \alpha.$$

Since $\rho_p = \text{diag}(p)$, the representation $\pi_\lambda(\rho_p)$ acts on this entire weight space as the scalar

$$p^\lambda \prod_{\alpha=(i,j)} \left(\frac{p_j}{p_i} \right)^{r_\alpha} = p^\lambda \prod_{\alpha} q_{\alpha}^{r_\alpha}.$$

The scalar depends only on the total weight, so orthonormalization within a weight space does not change it.

It remains to normalize. The Weyl character formula gives

$$s_\lambda(p) = \frac{\det[p_i^{\lambda_j + d - j}]_{i,j}}{\det[p_i^{d-j}]_{i,j}}.$$

Because both $p_1 > \dots > p_d$ and $\lambda_1 > \dots > \lambda_d$ are separated at order N , the identity permutation is the unique exponentially dominant term in the numerator. Therefore

$$s_\lambda(p) = p^\lambda \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right)^{-1} (1 + o(1)). \quad (64)$$

On every fixed cutoff, $\rho_{p,\lambda}$ thus converges to the diagonal probabilities

$$\prod_{\alpha} (1 - q_{\alpha}) q_{\alpha}^{r_{\alpha}},$$

which are exactly the occupation probabilities of $\bigotimes_{\alpha} \tau_{q_{\alpha}}$.

To pass from cutoff convergence to trace-norm convergence, choose L so that the limiting product thermal state has mass at least $1 - \varepsilon$ on total excitation $|\mathbf{r}| \leq L$. The finite-sector cutoff mass converges to the same number, hence is at least $1 - 2\varepsilon$ for large N . On the finite-dimensional cutoff, entrywise convergence implies trace-norm convergence. Compressing both states to the cutoff and using the gentle-measurement bound controls the discarded tails by $O(\sqrt{\varepsilon})$. Letting $\varepsilon \rightarrow 0$ proves (63). $\square$

The PRV inclusion becomes a beam splitter

Proposition 4.8 (Thermal PRV sector limit). *Let $\mu_n \vdash n$, $\nu_n \vdash m_n$, with $m_n/n \rightarrow \gamma > 1$, satisfy*

$$\mu_n/n \rightarrow p, \quad \nu_n/m_n \rightarrow p.$$

Set $\omega_n = \nu_n - \mu_n$. Then ω_n is dominant for all large n , and

$$\mathcal{C}_{\mu_n \rightarrow \nu_n}^{\text{PRV}}(\rho_{p,\mu_n}) \longrightarrow \bigotimes_{\alpha \in R_+} \tau_{q_{\alpha}^{(\gamma)}} \quad (65)$$

in trace norm, where

$$q_{\alpha}^{(\gamma)} = 1 - \frac{1 - q_{\alpha}}{\gamma}.$$

Consequently,

$$F\left(\mathcal{C}_{\mu_n \rightarrow \nu_n}^{\text{PRV}}(\rho_{p,\mu_n}), \rho_{p,\nu_n}\right) \longrightarrow F_{\text{orb}}(\gamma, p). \quad (66)$$

The convergence is uniform when p and the normalized weights range over compact subsets of the strict Weyl chamber.

Proof. Dominance follows from

$$\omega_{n,i} - \omega_{n,i+1} = (m_n - n)(p_i - p_{i+1}) + o(n) > 0.$$

Let $V = V_{\mu,\omega}$. Because V intertwines the $U(d)$ representations,

$$(\text{d}\pi_{\mu}(Y) \otimes I + I \otimes \text{d}\pi_{\omega}(Y))V = V \text{d}\pi_{\nu}(Y) \quad (67)$$

for every $Y \in \mathfrak{gl}_d$. Taking $Y = F_{\alpha}$ gives the exact identity

$$VF_{\alpha}^{(\nu)} = (F_{\alpha}^{(\mu)} \otimes I + I \otimes F_{\alpha}^{(\omega)})V. \quad (68)$$

Define

$$\eta_{\alpha}^{(n)} = \frac{\Delta_{\alpha}^{\mu}}{\Delta_{\alpha}^{\nu}}.$$

Since

$$\Delta_\alpha^\nu = \Delta_\alpha^\mu + \Delta_\alpha^\omega$$

and the normalized weights converge to p ,

$$\eta_\alpha^{(n)} \longrightarrow \frac{1}{\gamma}.$$

Apply (68) repeatedly to the normalized PBW monomials. Reordering into the fixed PBW order produces commutators, but Lemmas 4.4 and 4.6 show that all such terms are $o_L(1)$ on a fixed cutoff. Hence

$$Ve_{\mathbf{r}}^\nu = \sum_{\mathbf{a} \leq \mathbf{r}} \prod_{\alpha} \sqrt{\binom{r_\alpha}{a_\alpha}} (\eta_\alpha^{(n)})^{a_\alpha} (1 - \eta_\alpha^{(n)})^{r_\alpha - a_\alpha} e_{\mathbf{a}}^\mu \otimes e_{\mathbf{r}-\mathbf{a}}^\omega + o_L(1). \quad (69)$$

This is precisely the number-basis action of independent beam splitters with transmissivities $\eta_\alpha^{(n)}$.

Let

$$\sigma_{\mu,\nu} = \mathcal{C}_{\mu \rightarrow \nu}^{\text{PRV}}(\rho_{p,\mu}).$$

For fixed $\mathbf{r}, \mathbf{s}$,

$$\langle e_{\mathbf{r}}^\nu, \sigma_{\mu,\nu} e_{\mathbf{s}}^\nu \rangle = \frac{d_\mu}{d_\nu} \langle Ve_{\mathbf{r}}^\nu, (\rho_{p,\mu} \otimes I) Ve_{\mathbf{s}}^\nu \rangle.$$

Use (69) and Lemma 4.7. Off-diagonal entries vanish. For $\mathbf{r} = \mathbf{s}$, the Weyl dimension formula

$$d_\lambda = \prod_{i < j} \frac{\lambda_i - \lambda_j + j - i}{j - i}$$

gives

$$\frac{d_\mu}{d_\nu} = \prod_{\alpha} \eta_\alpha^{(n)} + o(1).$$

The limiting diagonal entry is therefore

$$\begin{aligned} & \prod_{\alpha} \eta_\alpha (1 - q_\alpha) \sum_{a=0}^{r_\alpha} \binom{r_\alpha}{a} (\eta_\alpha q_\alpha)^a (1 - \eta_\alpha)^{r_\alpha - a} \\ &= \prod_{\alpha} \eta_\alpha (1 - q_\alpha) (1 - \eta_\alpha + \eta_\alpha q_\alpha)^{r_\alpha}. \end{aligned}$$

Set

$$q'_\alpha = 1 - \eta_\alpha (1 - q_\alpha).$$

Then the last expression is

$$\prod_{\alpha} (1 - q'_\alpha) (q'_\alpha)^{r_\alpha}.$$

Since $\eta_\alpha = 1/\gamma$, this is the product thermal law with

$$q'_\alpha = q_\alpha^{(\gamma)}.$$

As in Lemma 4.7, convergence on every fixed cutoff implies tightness: choose L so that the limiting product thermal state has cutoff mass at least $1 - \varepsilon$; the corresponding cutoff mass of $\sigma_{\mu,\nu}$ converges to it. Finite-dimensional trace-norm convergence on the cutoff and the gentle-measurement bound then yield full trace-norm convergence. Finally, combine (65) with Lemma 4.7 and the thermal fidelity formula (23) to obtain (66). $\square$

Remark 4.9 (Stronger displaced-state interpretation). *The same contraction also implies, after tracking local unitary displacements, that the PRV channel converges on local displaced states to the quantum-limited amplifier. The global-fidelity achievability theorem does not require this stronger statement because covariance allows us to evaluate at the diagonal representative $g = I$.*

4.3 Known-spectrum achievability

Theorem 4.10 (Known-spectrum achievability). *The Schur-PRV cloner satisfies*

$$\liminf_{n \rightarrow \infty} \inf_{g \in \mathbf{U}(d)} F\left(\mathcal{C}_{n \rightarrow m}^{\text{PRV},p}(\rho_p^{\otimes n}), \rho_{p,g}^{\otimes m}\right) \geq F_{\text{orb}}(\gamma, p). \quad (70)$$

Proof. The channel is $\mathbf{U}(d)$ -covariant, as are the target states. Hence the fidelity in (70) is independent of g , and it is enough to take $g = I$.

Let

$$J_n(\mu, \nu) = P_{n,p}(\mu) K_n(\nu|\mu)$$

be the joint law of the measured input diagram and the sampled output diagram, and let

$$Q_m(\nu) = \sum_{\mu} J_n(\mu, \nu)$$

be its output marginal. The output and target states are block diagonal in the output Schur label:

$$\begin{aligned} \mathcal{C}_{n \rightarrow m}^{\text{PRV},p}(\rho_p^{\otimes n}) &= \bigoplus_{\nu} Q_m(\nu) \bar{\sigma}_{\nu} \otimes \frac{I_{\mathcal{S}_{\nu}}}{\dim \mathcal{S}_{\nu}}, \\ \rho_p^{\otimes m} &= \bigoplus_{\nu} P_{m,p}(\nu) \rho_{p,\nu} \otimes \frac{I_{\mathcal{S}_{\nu}}}{\dim \mathcal{S}_{\nu}}, \end{aligned}$$

where

$$\bar{\sigma}_{\nu} = \sum_{\mu} \frac{J_n(\mu, \nu)}{Q_m(\nu)} \mathcal{C}_{\mu \rightarrow \nu}^{\text{PRV}}(\rho_{p,\mu})$$

when $Q_m(\nu) > 0$. The Specht factors are identical, so they contribute fidelity one.

Take the good-pair set

$$G_n = \mathbf{T}_n(p) \times \mathbf{T}_m(p).$$

By Lemma 4.1,

$$J_n(G_n^c) = o(1) \quad \text{and} \quad \|Q_m - P_{m,p}\|_{\text{TV}} = o(1).$$

The second relation implies

$$B(Q_m, P_{m,p}) \longrightarrow 1,$$

because $1 - B(P, Q) \leq \|P - Q\|_{\text{TV}}$.

For every $(\mu, \nu) \in G_n$, the pair is admissible for all sufficiently large n , and Proposition 4.8 gives

$$B\left(\mathcal{C}_{\mu \rightarrow \nu}^{\text{PRV}}(\rho_{p,\mu}), \rho_{p,\nu}\right) \longrightarrow B_{\text{orb}}(\gamma, p)$$

uniformly over the good set. We can therefore apply Lemma 2.2 with

$$b = B_{\text{orb}}(\gamma, p), \quad P_n = P_{m,p}, \quad Q_n = Q_m.$$

It yields

$$\liminf_{n \rightarrow \infty} B\left(\mathcal{C}_{n \rightarrow m}^{\text{PRV},p}(\rho_p^{\otimes n}), \rho_p^{\otimes m}\right) \geq B_{\text{orb}}(\gamma, p).$$

Squaring proves (70). □

4.4 Known-spectrum global converse

Theorem 4.11 (Known-spectrum global converse). *For arbitrary channels*

$$\mathcal{M}_n : \mathcal{B}((\mathbb{C}^d)^{\otimes n}) \rightarrow \mathcal{B}((\mathbb{C}^d)^{\otimes m_n}),$$

$$\limsup_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{g \in \mathcal{U}(d)} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) \leq F_{\text{orb}}(\gamma, p). \quad (71)$$

Proof. It is enough to prove the corresponding root-fidelity bound. If the squared-fidelity statement failed, continuity of $x \mapsto \sqrt{x}$ on $[0, 1]$ would give $\varepsilon > 0$, a subsequence, and channels $\mathcal{M}_n$ such that

$$\inf_{g \in \mathcal{U}(d)} B(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) \geq B_{\text{orb}}(\gamma, p) + 3\varepsilon. \quad (72)$$

By the Gaussian converse (47), choose a finite cube $C_L \subset \mathbb{C}^R$ so large that

$$\sup_{\Lambda} \int_{C_L} B(\Lambda(\Phi_z), \Psi_z) \frac{dz}{|C_L|} \leq B_{\text{orb}}(\gamma, p) + \varepsilon. \quad (73)$$

The local orbit points

$$g_{n,z} = \exp(X_p(z)/\sqrt{n}), \quad z \in C_L,$$

belong to the original orbit, so (72) applies to each of them.

Let S_n be the inverse input LAN channel and T_{m_n} the forward output LAN channel from Theorem 3.4. Define

$$\Gamma_n = T_{m_n} \circ \mathcal{M}_n \circ S_n.$$

Apply Lemma 3.7 on the fixed compact set C_L . Uniformly for $z \in C_L$,

$$B(\Gamma_n(\Phi_z), \Psi_z) \geq B(\mathcal{M}_n(\rho_{p,g_{n,z}}^{\otimes n}), \rho_{p,g_{n,z}}^{\otimes m_n}) - o(1).$$

For all sufficiently large n , the right-hand side is at least

$$B_{\text{orb}}(\gamma, p) + 2\varepsilon.$$

Averaging over C_L therefore gives

$$\int_{C_L} B(\Gamma_n(\Phi_z), \Psi_z) \frac{dz}{|C_L|} \geq B_{\text{orb}}(\gamma, p) + 2\varepsilon,$$

contradicting (73), since Γ_n is an admissible oscillator channel. This proves the root-fidelity upper bound, and squaring gives (71). $\square$

Combining Theorems 4.10 and 4.11 gives the known-spectrum optimality theorem.

Corollary 4.12 (Known-spectrum asymptotic optimum).

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{g \in \mathcal{U}(d)} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = F_{\text{orb}}(\gamma, p),$$

and the value is achieved by the Schur-PRV cloner.

5 Unknown-spectrum adaptive cloning

We now assume the spectrum is not known. The model is the full nondegenerate family

$$\rho_{p,g} = U(g) \text{diag}(p) U(g)^*, \quad p \in \Delta_d^\circ.$$

The sharp statement is local at a fixed p , uniformly on compact subsets of the nondegenerate chamber. The additional loss relative to known spectrum comes from the classical LAN component.

5.1 Adaptive Schur–PRV protocol

The protocol again begins with the Schur transform and a measurement of $\mu \vdash n$. Now p is not known, so one does not sample ν from $P_{m,p}$. Instead choose a rounded dilation

$$R_{n,m}(\mu) = \nu \vdash m, \quad \left\| \nu - \frac{m}{n} \mu \right\|_\infty \leq C_d, \quad (74)$$

with $\nu_1 \geq \dots \geq \nu_d \geq 0$. Such a rounding map exists by distributing the integer rounding errors while preserving the order. If μ/n lies in a compact nondegenerate chamber and $\gamma > 1$, then $\omega = \nu - \mu$ is dominant for all large n , because

$$(\omega_i - \omega_{i+1}) = (\gamma_n - 1)(\mu_i - \mu_{i+1}) + O(1) > 0.$$

Then apply the same sector PRV channel $\mathcal{C}_{\mu \rightarrow \nu}^{\text{PRV}}$, prepare $I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu$, and invert the Schur transform. Denote the adaptive channel by $\mathcal{C}_{n \rightarrow m}^{\text{adPRV}}$.

The important point is that the PRV step itself does not need p . It only needs the measured sector μ and the chosen output sector ν .

5.2 The classical Young-label factor

Let $\mathbf{H} = \{x \in \mathbb{R}^d : \sum_i x_i = 0\}$ and $V_p = \text{diag}(p) - pp^T$ restricted to $\mathbf{H}$. The Schur label satisfies a central-limit theorem compatible with [Theorem 3.5](#):

$$X_n := \frac{\mu - np}{\sqrt{n}} \Rightarrow X \sim \mathbf{N}_{0,V_p}. \quad (75)$$

For a local spectrum perturbation $p + h/\sqrt{n}$, the limit is $\mathbf{N}_{h,V_p}$.

The adaptive dilation $\nu = R_{n,m}(\mu)$ gives

$$\frac{\nu - mp}{\sqrt{m}} = \frac{(m/n)\mu - mp}{\sqrt{m}} + o(1) = \sqrt{\frac{m}{n}} \frac{\mu - np}{\sqrt{n}} + o(1) \Rightarrow \mathbf{N}_{0,\gamma V_p}. \quad (76)$$

For the target m -copy state, the Young label has limit $\mathbf{N}_{0,V_p}$. Thus the classical squared fidelity contribution is

$$F_{\text{cl}}(\gamma, d) = F(\mathbf{N}_{0,\gamma V_p}, \mathbf{N}_{0,V_p}). \quad (77)$$

Lemma 5.1 (Classical Gaussian factor). *For any positive definite covariance matrix V on a k -dimensional real vector space,*

$$F(\mathbf{N}_{0,\gamma V}, \mathbf{N}_{0,V}) = \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^k. \quad (78)$$

In the spectrum problem $k = d - 1$, so (77) equals

$$F_{\text{cl}}(\gamma, d) = \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^{d-1}. \quad (79)$$

Proof. The squared Hellinger fidelity between centered Gaussians with covariances Σ_1, Σ_2 is

$$F = \frac{\det(\Sigma_1)^{1/2} \det(\Sigma_2)^{1/2}}{\det((\Sigma_1 + \Sigma_2)/2)}.$$

Taking $\Sigma_1 = \gamma V$, $\Sigma_2 = V$, and $\dim V = k$, gives

$$F = \frac{\gamma^{k/2} \det V}{((1 + \gamma)/2)^k \det V} = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^k.$$

□

Lemma 5.2 (Uniform adaptive Young-label factor). *Fix $\kappa > 0$, $L < \infty$, and $m_n/n \rightarrow \gamma > 1$. For $p \in \Delta_{d,\kappa}$ and $h \in \mathbf{H}$ with $\|h\| \leq L$, let μ be the Schur label drawn from $P_{n,p+h/\sqrt{n}}$, and let $\nu = R_{n,m}(\mu)$ be the rounded dilation (74). Then the Young-label experiment converges uniformly to the classical Gaussian amplification experiment*

$$X \sim \mathbf{N}_{h,V_p}, \quad Y = \sqrt{\gamma}X \sim \mathbf{N}_{\sqrt{\gamma}h,\gamma V_p},$$

whereas the target m -copy Young-label experiment converges uniformly to $\mathbf{N}_{\sqrt{\gamma}h,V_p}$. Consequently the Young-label contribution to squared fidelity converges uniformly to

$$F_{\text{cl}}(\gamma, d) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{d-1}.$$

Proof. The Young diagram is the classical component of the full LAN model in Theorem 3.5. Concretely, after centering and scaling,

$$X_n = \frac{\mu - np}{\sqrt{n}}$$

takes values on a lattice in $\mathbf{H}$. The classical part of compact-uniform LAN implies a local central limit theorem: after replacing each lattice point by its fundamental cell, the corresponding piecewise-constant density converges in $L^1(\mathbf{H})$, uniformly in $p \in \Delta_{d,\kappa}$ and $\|h\| \leq L$, to the density of $\mathbf{N}_{h,V_p}$.

The rounded dilation obeys

$$R_{n,m}(\mu) = \frac{m}{n}\mu + \delta_n(\mu), \quad \|\delta_n(\mu)\|_\infty \leq C_d.$$

Hence

$$\begin{aligned} \frac{R_{n,m}(\mu) - mp}{\sqrt{m}} &= \sqrt{\frac{m}{n}} \frac{\mu - np}{\sqrt{n}} + \frac{\delta_n(\mu)}{\sqrt{m}} \\ &= \sqrt{\frac{m}{n}} X_n + o(1), \end{aligned}$$

uniformly in μ . Under the local spectrum $p + h/\sqrt{n}$, the limiting mean is therefore $\sqrt{\gamma}h$, and the limiting covariance is γV_p . Thus the adaptive output-label experiment converges in total variation to

$$\mathbf{N}_{\sqrt{\gamma}h,\gamma V_p}.$$

The target m -copy Young label converges to

$$\mathbf{N}_{\sqrt{\gamma}h,V_p}.$$

If probability densities f_n, g_n converge in L^1 to f, g , then

$$\int |\sqrt{f_n} - \sqrt{f}|^2 \leq \|f_n - f\|_1,$$

and similarly for g_n . Their Hellinger affinities therefore converge:

$$\int \sqrt{f_n g_n} \longrightarrow \int \sqrt{f g}.$$

Applying this to the two rescaled Young-label distributions gives the uniform limit

$$B_{\text{labels}} \longrightarrow B\left(\mathbf{N}_{\sqrt{\gamma}h, \gamma V_p}, \mathbf{N}_{\sqrt{\gamma}h, V_p}\right) = B_{\text{cl}}(\gamma, d).$$

Squaring and using [Lemma 5.1](#) gives the stated factor $F_{\text{cl}}(\gamma, d)$. $\square$

A tempting but suboptimal alternative is to set $\hat{p} = \mu/n$ and then sample $\nu \sim P_{m, \hat{p}}$. At LAN scale this gives

$$\frac{\nu - mp}{\sqrt{m}} \approx \sqrt{\gamma}X + Y_0, \quad Y_0 \sim \mathbf{N}_{0, V_p},$$

so the covariance is $(\gamma + 1)V_p$, not γV_p . Its classical factor would be $(2\sqrt{\gamma + 1}/(\gamma + 2))^{d-1}$, which is worse.

5.3 Unknown-spectrum theorem

Define

$$F_{\text{univ}}(\gamma, p) = F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma}\right)^{d-1} \prod_{i < j} \frac{(1 - q_{ij})(1 - q_{ij}^{(\gamma)})}{(1 - \sqrt{q_{ij}q_{ij}^{(\gamma)}})^2}. \quad (80)$$

Lemma 5.3 (Variational form of root fidelity). *For faithful normal states ρ, τ ,*

$$B(\rho, \tau) = \inf_{A \geq 0} \frac{1}{2} \{\text{Tr}(\rho A) + \text{Tr}(\tau A^{-1})\}.$$

For commuting classical states this is the scalar identity $\sqrt{xy} = \inf_{a > 0} (ax + y/a)/2$ integrated over the sample space.

Proof. This is the standard operator arithmetic–geometric mean variational formula for Uhlmann root fidelity. In the commuting case it reduces pointwise to the scalar formula. $\square$

Proposition 5.4 (Classical Gaussian amplification converse). *For the classical Gaussian shift problem*

$$X \sim \mathbf{N}_{h, V_p} \longmapsto Y \sim \mathbf{N}_{\sqrt{\gamma}h, V_p}, \quad h \in \mathbf{H},$$

the flat-prior/minimax squared Hellinger fidelity is $F_{\text{cl}}(\gamma, d)$. It is achieved by the deterministic Markov kernel $Y = \sqrt{\gamma}X$.

Proof. The deterministic kernel $Y = \sqrt{\gamma}X$ gives output law $\mathbf{N}_{\sqrt{\gamma}h, \gamma V_p}$, whose squared Hellinger fidelity with the target $\mathbf{N}_{\sqrt{\gamma}h, V_p}$ is $F_{\text{cl}}(\gamma, d)$ by [Lemma 5.1](#). For the converse, the Hunt–Stein reduction of [Lemma 3.9](#) reduces the flat-prior problem to translation-covariant kernels. Such a kernel has the form

$$Y = \sqrt{\gamma}X + Z,$$

where $Z \sim \eta$ is independent of X . At $h = 0$, the output density is $f_\eta = \mathbf{N}_{0, \gamma V_p} * \eta$ and the target density is $g = \mathbf{N}_{0, V_p}$. Let $f_0 = \mathbf{N}_{0, \gamma V_p}$. Apply the variational formula of [Lemma 5.3](#) with the scalar witness

$$a_* = \sqrt{dg/df_0},$$

which is optimal for the pair (f_0, g) . Then

$$\int a_*^{-1} dg = \int \sqrt{f_0 g} = B_{\text{cl}}(\gamma, d).$$

The function a_* is an even Gaussian function and f_0 is an even Gaussian density. Therefore

$$r \mapsto \int a_*(y) f_0(y - r) dy = (a_* * f_0)(r)$$

is a centered Gaussian in r , maximized at $r = 0$, with value

$$(a_* * f_0)(0) = \int \sqrt{f_0 g} = B_{\text{cl}}(\gamma, d).$$

Consequently

$$\int a_* df_\eta = \int_{\mathbf{H}} \eta(dr) (a_* * f_0)(r) \leq B_{\text{cl}}(\gamma, d).$$

Hence

$$B(f_\eta, g) \leq \frac{1}{2} \left\{ \int a_* df_\eta + \int a_*^{-1} dg \right\} \leq B_{\text{cl}}(\gamma, d).$$

Equality is attained by $\eta = \delta_0$, i.e. by $Y = \sqrt{\gamma}X$. Squaring gives the claimed value $F_{\text{cl}}(\gamma, d)$. $\square$

Theorem 5.5 (Hybrid classical–quantum Gaussian product converse). *Let*

$$\Theta_{h,z}^{\text{in}} = \mathbf{N}_{h, V_p} \otimes \Phi_z, \quad \Theta_{h,z}^{\text{tar}} = \mathbf{N}_{\sqrt{\gamma}h, V_p} \otimes \Psi_z.$$

For the expanding cubes $C_L \subset \mathbf{H} \times \mathbb{C}^R$ defined below,

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \int_{C_L} B(\Lambda(\Theta_{h,z}^{\text{in}}), \Theta_{h,z}^{\text{tar}}) \frac{dh dz}{|C_L|} = B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p). \quad (81)$$

Consequently the optimal squared flat-prior/minimax fidelity is

$$F_{\text{cl}}(\gamma, d) F_{\text{orb}}(\gamma, p).$$

The value is achieved by the product of the deterministic classical amplifier $X \mapsto \sqrt{\gamma}X$ and the product quantum-limited oscillator amplifier.

Proof. The product of the deterministic classical amplifier

$$X \mapsto \sqrt{\gamma}X$$

and the product quantum-limited oscillator amplifier achieves the lower bound. Its origin output is

$$\mathbf{N}_{0, \gamma V_p} \otimes \bigotimes_{\alpha} \tau_{q_{\alpha}^{(\gamma)}},$$

whose root fidelity with the target factors as

$$B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p).$$

For the upper bound, [Lemma 3.9](#) reduces the expanding flat-prior problem to exactly covariant hybrid channels. After phase averaging at the channel level, the oscillator branches are also phase covariant. The residual-instrument lemma in [Appendix A](#) then writes the origin output as

$$\Xi(dy) = \int_{\mathbf{H}} \mathbf{N}_{r,\gamma V_p}(dy) \mathcal{I}(dr)(\Phi_0),$$

where every $\mathcal{I}(dr)$ is a trace-nonincreasing covariant oscillator amplifier branch. The variational witness for the least-noise product output factors into a classical Gaussian function and a decreasing product photon-number weight. The classical factor is maximized at $r = 0$, while [Lemma 3.13](#) bounds each quantum branch by the least-noise amplifier. Integrating over the residual instrument gives

$$B(\Xi, \Theta) \leq B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p).$$

The full branchwise calculation is given in [Proposition A.2](#). This proves (81), and squaring gives the stated optimal fidelity. $\square$

Theorem 5.6 (Unknown-spectrum local achievability). *Fix a compact local parameter set $K \subseteq \mathbf{H} \times \mathbb{C}^R$. For $(h, z) \in K$, put*

$$p_{n,h} = p + h/\sqrt{n}, \quad g_{n,z} = \exp(X_p(z)/\sqrt{n}),$$

and define

$$\rho_{n,h,z} = (\rho_{p_{n,h},g_{n,z}})^{\otimes n}, \quad \sigma_{m,h,z} = (\rho_{p_{n,h},g_{n,z}})^{\otimes m_n}.$$

Then the adaptive Schur-PRV channel satisfies

$$\liminf_{n \rightarrow \infty} \inf_{(h,z) \in K} F\left(\mathcal{C}_{n \rightarrow m}^{\text{adPRV}}(\rho_{n,h,z}), \sigma_{m,h,z}\right) \geq F_{\text{univ}}(\gamma, p). \quad (82)$$

Proof. Fix the compact set K . Since p is in the strict chamber, all local spectra

$$p_{n,h} = p + h/\sqrt{n}, \quad (h, z) \in K,$$

remain in a common compact strict chamber for sufficiently large n .

First remove the orbit coordinate. The adaptive channel is $U(d)$ -covariant, and the target transforms under the same representation. Therefore, for each fixed h , the fidelity is independent of z . We may set $z = 0$ and work with the diagonal state $\rho_{p_{n,h}}$.

Let $P_{n,h}(\mu)$ denote the input Young law at spectrum $p_{n,h}$. The adaptive protocol chooses

$$\nu = R_{n,m}(\mu)$$

deterministically, so its joint label law is

$$J_{n,h}(\mu, \nu) = P_{n,h}(\mu) \mathbf{1}_{\{\nu = R_{n,m}(\mu)\}}.$$

Let $Q_{n,h}$ be the resulting output-label law and let $P_{m,h}^{\text{tar}}$ be the Young law of the ideal m_n -copy target.

By [Lemma 5.2](#),

$$\inf_{(h,z) \in K} B(Q_{n,h}, P_{m,h}^{\text{tar}}) \longrightarrow B_{\text{cl}}(\gamma, d). \quad (83)$$

Now define the good set by

$$G_{n,h} = \{(\mu, \nu) : \nu = R_{n,m}(\mu), \mu \in \mathbb{T}_n(p_{n,h}), \nu \in \mathbb{T}_m(p_{n,h})\}.$$

Uniform Young concentration gives

$$\sup_{(h,z) \in K} J_{n,h}(G_{n,h}^c) \longrightarrow 0.$$

Indeed, if μ/n is within $n^{-1/3}$ of $p_{n,h}$, then

$$\frac{R_{n,m}(\mu)}{m} = \frac{\mu}{n} + O(1/m),$$

so the output diagram is also typical, after an inessential enlargement of the typical window. The same gap estimate as in [Lemma 4.1](#) shows that

$$\omega = R_{n,m}(\mu) - \mu$$

is dominant uniformly on the good set.

Conditional on a good pair (μ, ν) , the normalized weights satisfy

$$\frac{\mu}{n} \rightarrow p, \quad \frac{\nu}{m_n} \rightarrow p, \quad \frac{\Delta_{ij}^\nu}{\Delta_{ij}^\mu} \rightarrow \gamma$$

uniformly for $(h, z) \in K$. The thermal PRV sector theorem therefore gives

$$\inf_{\substack{(h,z) \in K \\ (\mu,\nu) \in G_{n,h}}} B\left(C_{\mu \rightarrow \nu}^{\text{PRV}}(\rho_{p_{n,h},\mu}), \rho_{p_{n,h},\nu}\right) \longrightarrow B_{\text{orb}}(\gamma, p). \quad (84)$$

Replacing $p_{n,h}$ by p in the limiting factor is valid uniformly because $p_{n,h} \rightarrow p$ and $B_{\text{orb}}(\gamma, \cdot)$ is continuous in the strict chamber.

The output and target have the same maximally mixed Specht factors. Apply [Lemma 2.2](#) to the label law $J_{n,h}$, the label affinity (83), and the conditional sector bound (84). Uniformly on K ,

$$\liminf_{n \rightarrow \infty} B\left(C_{n \rightarrow m}^{\text{adPRV}}(\rho_{n,h,z}), \sigma_{m,h,z}\right) \geq B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p).$$

Squaring gives (82). □

For the converse, a fixed compact local window is not least favorable: on a singleton one can prepare the target and get fidelity one. The correct local minimax converse uses expanding windows. Let

$$C_L = \{(h, z) \in \mathbf{H} \times \mathbb{C}^R : \|h\| \leq L, |\Re z_\alpha|, |\Im z_\alpha| \leq L \text{ for all } \alpha\}.$$

Theorem 5.7 (Unknown-spectrum expanding-window local converse). *For the full-rank nondegenerate LAN model of [Theorem 3.5](#), one has*

$$\lim_{L \rightarrow \infty} \limsup_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{(h,z) \in C_L} F(\mathcal{M}_n(\rho_{n,h,z}), \sigma_{m,h,z}) \leq F_{\text{univ}}(\gamma, p). \quad (85)$$

Together with [Theorem 5.6](#), this identifies $F_{\text{univ}}(\gamma, p)$ as the local asymptotic minimax value in the expanding-window LAN sense.

Proof. We prove the root-fidelity form. Define

$$v_{n,L} = \sup_{\mathcal{M}_n} \inf_{(h,z) \in C_L} B(\mathcal{M}_n(\rho_{n,h,z}), \sigma_{m,h,z}).$$

Suppose, contrary to the theorem, that

$$\lim_{L \rightarrow \infty} \limsup_{n \rightarrow \infty} v_{n,L} > B_{\text{univ}}(\gamma, p).$$

Then there is $\varepsilon > 0$ such that the left-hand side is at least

$$B_{\text{univ}}(\gamma, p) + 4\varepsilon.$$

By the hybrid Gaussian theorem (81), choose $L < \infty$ large enough that

$$\sup_{\Lambda} \int_{C_L} B\left(\Lambda(\Theta_{h,z}^{\text{in}}, \Theta_{h,z}^{\text{tar}})\right) \frac{dh dz}{|C_L|} \leq B_{\text{univ}}(\gamma, p) + \varepsilon. \quad (86)$$

For this fixed L , the definition of the limsup provides a subsequence, still denoted n , and channels $\mathcal{M}_n$ satisfying

$$\inf_{(h,z) \in C_L} B(\mathcal{M}_n(\rho_{n,h,z}), \sigma_{m,h,z}) \geq B_{\text{univ}}(\gamma, p) + 3\varepsilon. \quad (87)$$

If the supremum is not attained, choose a channel within $\varepsilon/2$ of it; this only changes the numerical margin.

Let S_n be the inverse input LAN channel and T_{m_n} the forward output LAN channel from Theorem 3.5. Set

$$\Gamma_n = T_{m_n} \circ \mathcal{M}_n \circ S_n.$$

Apply Lemma 3.7 on the fixed compact set C_L . Uniformly for $(h, z) \in C_L$,

$$B\left(\Gamma_n(\Theta_{h,z}^{\text{in}}, \Theta_{h,z}^{\text{tar}})\right) \geq B(\mathcal{M}_n(\rho_{n,h,z}), \sigma_{m,h,z}) - o(1).$$

For all sufficiently large n , (87) therefore implies

$$\inf_{(h,z) \in C_L} B\left(\Gamma_n(\Theta_{h,z}^{\text{in}}, \Theta_{h,z}^{\text{tar}})\right) \geq B_{\text{univ}}(\gamma, p) + 2\varepsilon.$$

Averaging this inequality over C_L contradicts (86). Hence the expanding-window root-fidelity value is at most $B_{\text{univ}}(\gamma, p)$. Squaring proves (85). $\square$

The expanding-window local converse implies a genuine global worst-case statement on compact subsets of the strict Weyl chamber. The needed uniformity is supplied by Lemmas 2.3, 4.6 and 5.2 and Corollary 3.6.

Corollary 5.8 (Compact-spectrum global worst-case value). *Fix $\kappa > 0$ and let*

$$\Delta_{d,\kappa} = \{p : p_d \geq \kappa, p_i - p_{i+1} \geq \kappa, \sum_i p_i = 1\}.$$

Then

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{\substack{p \in \Delta_{d,\kappa} \\ g \in \mathcal{U}(d)}} F\left(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) = \inf_{p \in \Delta_{d,\kappa}} F_{\text{univ}}(\gamma, p). \quad (88)$$

The adaptive Schur–PRV channel achieves the lower bound.

Proof. For achievability, use the same adaptive Schur–PRV channel for every spectrum. The Young concentration, compact-uniform LAN, PBW/Fock contraction, and adaptive label estimates are uniform on $\Delta_{d,\kappa}$ by

Lemmas 2.3, 4.6 and 5.2 and Corollary 3.6.

The proof of [Theorem 5.6](#) is therefore uniform in $p \in \Delta_{d,\kappa}$, giving

$$\liminf_{n \rightarrow \infty} \inf_{\substack{p \in \Delta_{d,\kappa} \\ g \in \mathcal{U}(d)}} F\left(\mathcal{C}_{n \rightarrow m}^{\text{adPRV}}(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) \geq \inf_{p \in \Delta_{d,\kappa}} F_{\text{univ}}(\gamma, p).$$

For the converse, suppose there are $\varepsilon > 0$, a subsequence, and channels $\mathcal{M}_n$ such that

$$\inf_{\substack{p \in \Delta_{d,\kappa} \\ g \in \mathcal{U}(d)}} F\left(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) \geq \inf_{p \in \Delta_{d,\kappa}} F_{\text{univ}}(\gamma, p) + 4\varepsilon. \quad (89)$$

The relative interior of $\Delta_{d,\kappa}$ is dense in $\Delta_{d,\kappa}$, and $F_{\text{univ}}(\gamma, p)$ is continuous there. Choose an interior point p_0 such that

$$F_{\text{univ}}(\gamma, p_0) \leq \inf_{p \in \Delta_{d,\kappa}} F_{\text{univ}}(\gamma, p) + \varepsilon.$$

Because p_0 has positive distance from the boundary of $\Delta_{d,\kappa}$, for every fixed L all local perturbations

$$p_0 + h/\sqrt{n}, \quad (h, z) \in C_L,$$

remain in $\Delta_{d,\kappa}$ for sufficiently large n . The global bound (89) then implies, on every fixed local cube,

$$F\left(\mathcal{M}_n(\rho_{n,h,z}^{(p_0)}), \sigma_{m,h,z}^{(p_0)}\right) \geq F_{\text{univ}}(\gamma, p_0) + 3\varepsilon,$$

where the superscript indicates that the local model is centered at p_0 . Taking the expanding-window limit contradicts [Theorem 5.7](#) at p_0 . This proves the upper bound and hence (88). $\square$

A global worst-case statement over the entire full-rank simplex without a gap condition is not covered by (88): as p approaches degeneracy or the boundary, the LAN model changes and the formula $F_{\text{univ}}(\gamma, p)$ need not remain the correct local value.

Remark 5.9 (Known side information at intermediate precision). *If the spectrum is supplied externally to accuracy $o(n^{-1/2})$, the classical penalty disappears and the known-spectrum value F_{orb} is recovered. If the supplied accuracy is only $O(n^{-1/2})$, the classical LAN problem changes by replacing the input covariance V_p with the posterior covariance after incorporating that side information.*

6 Comparison with existing mixed-state cloning and broadcasting results

Broadcasting versus cloning. The present work is about cloning in the global product sense: the target is $\rho^{\otimes m}$. Broadcasting is weaker: it requires only that selected output marginals reproduce the input state. A perfect broadcaster can have strong correlations among the outputs, and therefore may have poor global fidelity with $\rho^{\otimes m}$. This distinction is essential for interpreting the mixed-state literature. The single-copy no-broadcasting theorem rules out exact broadcasting of noncommuting families from one input copy [\[BCFJS96\]](#). With several input copies, however, exact marginal broadcasting can be possible. In particular, [\[DMP05\]](#) showed that the no-broadcasting theorem does

not extend directly to more than one input copy for mixed qubits and introduced superbroadcasting, where the output one-site marginals can even be purer than the input for sufficiently many input copies. Such purifying marginal maps can be degraded by local depolarization to exact broadcasting, and exact broadcasting was also constructed explicitly for displaced thermal oscillator states with known thermal photon number in [DPS06]. The review/extension [BDMP05] states explicitly that for sufficiently many input copies ideal broadcasting of noncommuting mixed states is possible, and the universal and phase-covariant maps are developed further in [BDMP06]. These results are not contradicted by no-cloning, because they do not produce an uncorrelated product $\rho^{\otimes m}$.

Earlier mixed-state cloning bounds. Rastegin studied global-fidelity, relative-error, and state-dependent limits for mixed-state cloning [Ras03a, Ras02, Ras03c]. Fan and collaborators constructed mixed-state cloners in symmetric subspaces and for identical mixed qubits [Fan03, FLS07]; Kazakov studied partial cloning and mixed-qubit copying in another restricted setting [Kaz07]. These works are useful benchmarks, but they do not provide a sharp arbitrary-dimensional proportional-output all-site asymptotic formula. The Gaussian setting is an exception in infinite dimension: Guta–Matsumoto solved displaced mixed Gaussian cloning via least-noise amplification [GM06]. Our finite-dimensional converse reduces to that theorem through LAN.

Other cloning models. Several other cloning-related settings are operationally distinct from the global product-fidelity problem considered here. Examples include probabilistic cloning [DG98], economical, phase-covariant, asymmetric, and telecloning models [NG98, BCDM00, Cer00, MJPV99], transformation cloning and superreplication [CDP08, CYY13, GCMBC14, DSS15, CYH15, CY16], channel cloning [SGKS25], and recent limitation or structural results for cloning maps [HT24, BGSQ26]. They provide useful context, but they do not identify the mixed-state $\rho^{\otimes n} \mapsto \rho^{\otimes m}$ global-fidelity asymptotics addressed in this paper.

Random purification plus pure-state cloning. The coherent-inference construction of [LTHC26a] treats rank- r mixed states by first applying a random-purification channel and then the optimal pure-state cloner in dimension dr . Tracing out the purifying registers gives mixed-state clones. Their stated all-site guarantee has the pure-state-cloning form

$$F_{\text{RP}}(n, m; d, r) \geq \frac{\binom{n+dr-1}{dr-1}}{\binom{m+dr-1}{dr-1}}. \quad (90)$$

With $m/n \rightarrow \gamma$,

$$\frac{\binom{n+dr-1}{dr-1}}{\binom{m+dr-1}{dr-1}} = \prod_{a=1}^{dr-1} \frac{n+a}{m+a} \longrightarrow \gamma^{-(dr-1)}. \quad (91)$$

For full-rank states, $r = d$, this is $\gamma^{-(d^2-1)}$.

Our full-rank fixed-spectrum orbit problem has

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} f_{\gamma}(q_{ij}),$$

where

$$f_{\gamma}(q) := \left(\frac{\sqrt{\gamma} + \sqrt{q(\gamma - 1 + q)}}{\gamma + q} \right)^2.$$

We now claim that for every $q > 0$ and $\gamma > 1$,

$$f_\gamma(q) > \frac{1}{\gamma}. \quad (92)$$

Let us first check why each one-root factor is larger than $1/\gamma$. Since the denominator $\gamma + q$ is positive, the comparison $f_\gamma(q) > 1/\gamma$ is equivalent, after taking positive square roots, to

$$\sqrt{\gamma} + \sqrt{q(\gamma - 1 + q)} > \frac{\gamma + q}{\sqrt{\gamma}} = \sqrt{\gamma} + \frac{q}{\sqrt{\gamma}}.$$

After cancelling $\sqrt{\gamma}$, this becomes

$$\sqrt{q(\gamma - 1 + q)} > \frac{q}{\sqrt{\gamma}}.$$

Both sides are nonnegative, and squaring gives

$$q(\gamma - 1 + q) > \frac{q^2}{\gamma},$$

or equivalently

$$q(\gamma - 1) \left(1 + \frac{q}{\gamma}\right) > 0.$$

This last inequality is strict for $q > 0$ and $\gamma > 1$. Hence, for every $q > 0$ and $\gamma > 1$, we have $f_\gamma(q) > \frac{1}{\gamma}$, as claimed.

Therefore, with $R = d(d - 1)/2$,

$$F_{\text{orb}}(\gamma, p) > \gamma^{-R}. \quad (93)$$

Since $R < d^2 - 1$ for $d \geq 2$,

$$F_{\text{orb}}(\gamma, p) > \gamma^{-R} > \gamma^{-(d^2-1)}. \quad (94)$$

Thus the known-spectrum Schur-PRV optimum is strictly larger than the full-rank random-purification guarantee for all $\gamma > 1$.

For unknown spectrum,

$$F_{\text{univ}}(\gamma, p) = F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p).$$

Since

$$\frac{2\sqrt{\gamma}}{1 + \gamma} > \gamma^{-1/2} \quad (\gamma > 1),$$

we have

$$F_{\text{univ}}(\gamma, p) > \gamma^{-(d-1)/2} \gamma^{-d(d-1)/2} = \gamma^{-(d^2-1)/2} > \gamma^{-(d^2-1)}. \quad (95)$$

Again, this compares our sharp local full-rank asymptotics to the universal purify-and-clone guarantee, not to a proven exact fidelity formula for that specific channel restricted to a fixed orbit. The companion QPA work treats coherent amplification of eigenstates rather than direct mixed-state product cloning, and is therefore a related purification benchmark rather than a substitute for the comparison above [LTHC26b].

6.1 Large-gamma and near-identity expansions

The expansion in powers of $1/\gamma$ is only a way to compare high expansion ratios. It is not used in the asymptotic proof, where $\gamma > 1$ is fixed and the small parameter is $1/n$. When γ is close to one, the natural small parameter is $\epsilon = \gamma - 1$, and the exact formulas above should be used.

For fixed p , as $\gamma \rightarrow \infty$,

$$F_{\text{orb}}(\gamma, p) = \gamma^{-d(d-1)/2} \prod_{i < j} \left(1 + \sqrt{\frac{p_j}{p_i}}\right)^2 \left(1 + O(\gamma^{-1})\right). \quad (96)$$

The product prefactor matters for the sharp leading constant, but not for the exponent. The inverse ratio $(1 + \sqrt{p_i/p_j})^2$ would be incorrect unless accompanied by the missing factor p_j/p_i for each root.

Near $\gamma = 1$, write $\gamma = 1 + \epsilon$. For each root,

$$f_{1+\epsilon}(q) = 1 - \frac{\epsilon^2}{4q} + O(\epsilon^3),$$

so the known-spectrum infidelity is quadratic in $\gamma - 1$. By contrast, the pure-cloner dimension-ratio guarantee $\gamma^{-(d^2-1)}$ has a linear loss

$$(1 + \epsilon)^{-(d^2-1)} = 1 - (d^2 - 1)\epsilon + O(\epsilon^2).$$

7 Outlook and discussion

The results identify the asymptotic global fidelity for full-rank nondegenerate mixed-state cloning in the proportional regime. The known-spectrum problem is governed by the root-mode quantum amplifier, whereas the unknown-spectrum problem contains, in addition, a classical Gaussian amplification factor for the Young-label/spectral LAN component. The Schur-PRV construction realizes the quantum amplifier directly at the level of Weyl sectors, and the adaptive rounded-dilation rule for Young diagrams realizes the classical amplifier.

Several extensions remain natural. First, one can ask for quantitative finite- n estimates. This would require explicit constants in Keyl-Werner concentration, quantitative LAN bounds, and quantitative versions of the PBW/Fock approximation. Second, degenerate spectra and rank-deficient states require separate analysis: degeneracies introduce stabilizers and collapse some root modes, while boundary points of the simplex lead to non-Gaussian or constrained LAN limits. Third, it would be useful to understand whether the PRV sector map is exactly optimal at finite sector sizes, beyond its asymptotic optimality for the global cloning problem. Finally, implementation of the protocol requires efficient Schur transforms and efficient realization of the relevant Cartan/PRV intertwiners.

[Internal note: if targeting a shorter journal version, this outlook can be reduced to one paragraph. Circuit implementation, finite-sector optimality, and degenerate spectra are natural follow-up projects, not hypotheses of the present theorem.]

[Internal note: former appendices A–C have been incorporated into the main text. The proof-status appendix has been removed; its role is now only this removable comment. The theorem statements are asymptotic and do not claim finite- n rates or boundary/degenerate-spectrum formulas.]

A The hybrid product converse: residual instruments and conic order

This appendix proves the hybrid classical–quantum Gaussian product converse used in [Theorem 5.5](#). The main issue is that a channel could correlate its classical spectral error with the quantum amplifier noise. We show that product covariance forces a much simpler structure: the classical output is the correctly amplified Gaussian input plus an additive residual r , and, conditional on that residual, the oscillator output is a covariant trace-nonincreasing amplifier branch. The classical and quantum witness bounds can then be applied branch by branch.

We first recall the language used in the proof. A classical–quantum output with classical value $y \in \mathsf{H}$ is represented by a positive operator-valued measure

$$E \longmapsto \Xi(E)$$

on Borel sets $E \subseteq \mathsf{H}$. Its trace $\text{Tr} \Xi(E)$ is the probability that the classical output lies in E , and $\Xi(E)/\text{Tr} \Xi(E)$, when defined, is the corresponding conditional quantum state.

A *quantum instrument* is a countably additive assignment

$$E \longmapsto \mathcal{I}_E$$

of completely positive trace-nonincreasing maps such that $\mathcal{I}_{\mathsf{H}}$ is trace preserving. It simultaneously describes a classical outcome and the conditional quantum output. The notation $\mathcal{I}(dr)$ below is the usual measure notation for this Borel instrument; all identities are understood after integration over Borel sets or against bounded measurable test functions.

The proof combines the Hunt–Stein reduction from [Lemma 3.9](#), the residual-instrument decomposition below, the variational formula [Lemma 5.3](#), and the tensorized conic order [Lemma 3.13](#).

Lemma A.1 (Residual-instrument form of covariant hybrid channels). *Let Λ be a normal channel from the hybrid classical–quantum input model to the hybrid output model, exactly covariant under $G = \mathsf{H} \times \mathbb{C}^R$. The input action translates the classical variable by s and displaces the oscillator by w ; the output action translates the classical variable by $\sqrt{\gamma}s$ and displaces the oscillator by $\sqrt{\gamma}w$. Then there is a Borel quantum instrument $E \mapsto \mathcal{I}_E$ on H , whose values are trace-nonincreasing CP maps on the oscillators and are displacement covariant with gain $\sqrt{\gamma}$, such that for every bounded measurable test function a and every bounded oscillator observable M , the output Ξ of Λ at the centered input $\mathsf{N}_{0,V_p} \otimes \Phi_0$ satisfies*

$$\int a(y) \text{Tr}[M \Xi(dy)] = \int_{\mathsf{H}} \left(\int a(y) \mathsf{N}_{r,\gamma V_p}(dy) \right) \text{Tr}[M \mathcal{I}(dr)(\Phi_0)]. \quad (97)$$

Equivalently, in measure notation,

$$\Xi(dy) = \int_{\mathsf{H}} \mathsf{N}_{r,\gamma V_p}(dy) \mathcal{I}(dr)(\Phi_0), \quad \int_{\mathsf{H}} \text{Tr}[\mathcal{I}(dr)(\Phi_0)] = 1. \quad (98)$$

Proof. We use the standard classical–quantum Markov-kernel formulation. For a classical input point $x \in \mathsf{H}$, the channel induces a quantum instrument $E \mapsto \Lambda_E^x$ on the oscillator system, where $E \subseteq \mathsf{H}$ is a Borel output event. Product covariance gives, for all Borel E , classical shifts s , oscillator displacements w , and oscillator trace-class inputs X ,

$$\Lambda_{E+\sqrt{\gamma}s}^{x+s}(D(w)XD(w)^*) = D(\sqrt{\gamma}w)\Lambda_E^x(X)D(\sqrt{\gamma}w)^*. \quad (99)$$

The classical-shift subgroup acts trivially on the oscillator output; it only changes the classical outcome by $\sqrt{\gamma}s$. Setting $w = 0$, $x = 0$, and $s = x$, we obtain

$$\Lambda_E^x(X) = \Lambda_{E-\sqrt{\gamma}x}^0(X).$$

Define the residual instrument by $\mathcal{I}_E(X) := \Lambda_E^0(X)$. Then for a point input x , the output classical variable has the form $\sqrt{\gamma}x + r$, with residual instrument branch $\mathcal{I}(dr)$. Integrating the point-input identity against $x \sim \mathbf{N}_{0,V_p}$ gives (97), because $\sqrt{\gamma}X \sim \mathbf{N}_{0,\gamma V_p}$ and the residual branch is independent of the amplified Gaussian input.

Finally, setting $x = 0$ and using (99) with arbitrary w gives, for every Borel E ,

$$\mathcal{I}_E(D(w)XD(w)^*) = D(\sqrt{\gamma}w)\mathcal{I}_E(X)D(\sqrt{\gamma}w)^*.$$

Thus each Borel branch is displacement covariant with gain $\sqrt{\gamma}$. Trace preservation of Λ gives $\int \text{Tr}[\mathcal{I}(dr)(\Phi_0)] = \text{Tr} \Phi_0 = 1$. $\square$

Proposition A.2 (Hybrid converse from residual instruments). *The hybrid Gaussian product converse, Theorem 5.5, holds.*

Proof. By Lemma 3.9 for the amenable product group $\mathbf{H} \times \mathbb{C}^R$, it suffices to prove the origin bound for exactly product-covariant channels. We first average the channel, not merely its output, over the per-mode phase group $\mathbf{U}(1)^R$, conjugating oscillator input and output by $\bigotimes_{\alpha} R(\theta_{\alpha})$ and integrating over θ . This averaging preserves the product covariance and the gain $\sqrt{\gamma}$, because phase rotations satisfy

$$R(\theta)D(\sqrt{\gamma}w)R(\theta)^* = D(\sqrt{\gamma}e^{i\theta}w).$$

Since the thermal input and target are phase invariant and root fidelity is concave in the channel output, this channel twirl cannot decrease the origin fidelity. The resulting channel is phase covariant in every oscillator mode. Therefore the residual branches supplied by Lemma A.1 are both displacement covariant with gain $\sqrt{\gamma}$ and phase covariant, exactly as required by Lemma 3.13; in particular the oscillator output is diagonal in the joint number basis.

Let

$$\Theta = \mathbf{N}_{0,V_p} \otimes \bigotimes_{\alpha} \tau_{q_{\alpha}}, \quad \Omega_* = \mathbf{N}_{0,\gamma V_p} \otimes \bigotimes_{\alpha} \tau_{q_{\alpha}^{(\gamma)}}$$

be the target and least-noise product output at the origin. In the variational formula of Lemma 5.3, the optimizer for the commuting pair (Ω_*, Θ) factors as $A_*(y, \mathbf{k}) = a_0(y)W_{\mathbf{k}}$, where

$$a_0(y) = \sqrt{\frac{d\mathbf{N}_{0,V_p}}{d\mathbf{N}_{0,\gamma V_p}}}(y), \quad W_{\mathbf{k}} = C \prod_{\alpha} \left(\frac{q_{\alpha}}{q_{\alpha}^{(\gamma)}} \right)^{k_{\alpha}/2}.$$

The function a_0 is a centered Gaussian decreasing in $\|y\|_{V_p^{-1}}$, and W is a bounded product of decreasing photon-number weights.

Let Ξ be the output of an exactly covariant hybrid channel at the origin. By Lemma A.1,

$$\Xi(dy) = \int \mathbf{N}_{r,\gamma V_p}(dy) \mathcal{I}(dr)(\Phi_0).$$

For each residual r , the classical factor obeys

$$\int a_0(y) \mathbf{N}_{r,\gamma V_p}(dy) \leq \int a_0(y) \mathbf{N}_{0,\gamma V_p}(dy) = B_{\text{cl}}(\gamma, d),$$

since the left-hand side is a centered Gaussian function of r , maximized at $r = 0$. For the oscillator branch, [Lemma 3.13](#) gives

$$\mathrm{Tr}[\mathcal{I}(dr)(\Phi_0)W(N)] \leq \mathrm{Tr}[\mathcal{I}(dr)(\Phi_0)] B_{\mathrm{orb}}(\gamma, p).$$

Multiplying these two bounds and integrating over r yields

$$\mathrm{Tr}(\Xi A_*) \leq B_{\mathrm{cl}}(\gamma, d) B_{\mathrm{orb}}(\gamma, p).$$

The other variational term is fixed by construction,

$$\mathrm{Tr}(\Theta A_*^{-1}) = B_{\mathrm{cl}}(\gamma, d) B_{\mathrm{orb}}(\gamma, p).$$

Therefore

$$B(\Xi, \Theta) \leq B_{\mathrm{cl}}(\gamma, d) B_{\mathrm{orb}}(\gamma, p),$$

and squaring gives $F_{\mathrm{cl}} F_{\mathrm{orb}}$. The Hunt–Stein reduction gives the same bound for arbitrary channels in the flat-prior and expanding-window minimax limits. $\square$

Remark A.3 (What the residual form removes). *The possible obstruction was a classical residual correlated with a trace-decreasing quantum branch, analogous to a heralded amplifier. [Lemma A.1](#) shows that exact product covariance forces the amplified input Gaussian to be independent of the residual branch, and [Lemma 3.13](#) bounds each covariant oscillator branch. Hence correlations between classical residual noise and quantum amplifier noise cannot improve the product fidelity.*

[Internal reference note: Keyl–Werner supplies Young-frame spectrum concentration; Kahn–Guta supplies finite-dimensional LAN with classical Gaussian and displaced-thermal oscillator components; Guta–Matsumoto supplies mixed Gaussian cloning via amplification and stochastic ordering; Li–Theil–Harrow–Chuang supply the random-purification comparison and QPA context used in [Section 6](#). This note replaces the former “References to technical inputs” appendix and can be deleted once the bibliography and citation placement are finalized.]

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